Macroscopic Information Transfer in Adaptive Tape Transposition

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Abstract

We study the original executions of deterministic multitape machines that transpose a row-major matrix within linear time. An offline replay records selected run entries and incoming work symbols without fixing a fine head trace. For fixed resources, uniformly over every positive entry width, all but exponentially few ambient inputs on which the machine is timely and correct require two work heads to make linearly many macroscopic returns. Causal-fragment replay gives original input/output cut expansion and expansion to off-carrier return predecessors. A guarded projection from an existing native simulator also gives expansion from every large output set to original crossings, and capacity certificates on original entry relations, at canonical widths. These results constrain the actual computation without assuming that its original trace is shared by many inputs.

Further bounds measure incoming values that escape fixed local reconstructions, and changes to stored words relative to locally readable references. A dynamic-carrier argument forces two simultaneous linear-sized stores. Redundant archives and a cheap historical-feedback control show why large stores or extensive reuse alone do not prove a time lower bound. The replay, cut, getter, loader, block-program and actual simulator-transport arguments are Lean-verified. The dynamic-carrier, archive and physical monitor constructions have complete written proofs; their remaining Lean obligations and the consequences conditional on them are explicit. The question of unrestricted linear-time transposition remains open.

Contributions. Hans Brende chose the problem and directed the research. GPT-6 Astra (OpenAI) developed the mathematics, Lean formalizations and text under his direction.

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1 Introduction

The question (T) asks whether one fixed deterministic multitape Turing machine can transpose all square matrices in time linear in their row-major encoding length. The input is read-only; work tapes are read/write; output is append-only and cannot be read. A mathematical decoder may inspect records in any order, whereas the machine must obtain its operands with the actual tape heads.

The seventeenth paper identifies a gap between restrictions on decoding rules and restrictions on complete tape computations [1, Sections 3 and 7.4]. Its order and symmetry theorems apply to structured one-hop decoders. Obtaining such structure, with its advice and intermediate states paid for, from arbitrary native executions remains open. We work on the execution side of this gap: which records must carry information, where must that information pass, and how much can fixed local descriptions explain?

Original adaptive executions. Earlier macroscopic-crossing results constrain oblivious executions and sufficiently heavy original traces. Their general adaptive application uses a low-trace simulator whose dependency graph differs from the original machine's graph [3]. Our direct replay instead pays for selected intervals of the original execution.

Fix the number of work tapes, the alphabet and a linear time constant. There are positive constants $\varepsilon, \gamma, \eta$ such that fewer than $2^{(1-\eta)P}$ length-P inputs are both timely and correctly transposed and admit a free work head leaving at most γP returns of excursion greater than εP on the other heads. The threshold is uniform over all positive entry widths. Consequently two heads make linearly many macroscopic returns on every other timely correct input. The exception count is relative to the ambient cube; it is not a conditional proportion of an arbitrarily small success set.

The proof retains every visit to a spatial slab of the free head. Its memory persists through gaps between retained runs. On another tape, a first read outside the paid entry window either reads a virgin cell or has an actual long-return predecessor. One joint sparse event/value record supplies these symbols. The replayed correct outputs recover raw bits omitted from the message.

A second replay retains causal fragments rather than every step in a slab. It pays complete cut values and processes intervening writes chronologically. This gives simultaneous expansion for arbitrary macroscopic raw demand sets and, after localization, expansion to off-carrier long-return predecessors. The localized result concerns the output sets supplied by the slab geometry.

Two further conclusions use the existing aligned simulator of [4]. We prove an actual path and cone correspondence, paying guards for the changed clock, head positions and delayed outputs. It transfers expansion from *every* large output set to original crossings and gives storage-capacity certificates on the original entry relation. Their asymptotic statements use canonical widths. The finite eroded-input cut invoice and the direct replay results hold at every positive width. No common original trace is assumed in either route.

What stored data must do. A one-work-tape simulation handles runs in which at every configuration at most one work tape has more than m nonblank cells. The carrier may change arbitrarily often: sparse sketches and a lazily reset visited interval preserve the computation through a handoff. Combining the simulation with the one-head bound forces two simultaneous large stores. A linear-overhead redundant-archive wrapper shows why this conclusion does not imply independence of their contents.

The extensions then test more informative descriptions of those stores. Fixed getters reconstruct incoming values from continuous raw windows, paid observer state, one shared label and the strictly prior correct-output string. Shared spatial checkpoints supply such getters for deterministic

scans. Changing-carrier and historical-cache arguments compare whole words with fixed locally readable reference families and force long secondary modification episodes. Persistence estimates quantify marginal uncertainty, including uncertainty after conditioning on the emitted prefix. The historical-feedback appendix measures dated work-block reuse through actual halt. A cheap copying control bounds its interpretation: a large historical statistic alone does not account for the cost of transposition.

Proofs and formal status. Each numbered result identifies its formal status. The replay, original return and cut bounds, coherent-cluster extension, getters, paid loader, complete block-program lemma and actual simulator transport are proved in Lean. The carrier, catalogue, eraser, archive and physical monitor constructions have complete written proofs but retain native formalization obligations. Their checked consequences explicitly assume the missing construction bounds. No additional axiom supplies them. The Lean declarations are authoritative; the English outlines are reading aids. Appendix J states the precise boundary.

Relation to earlier work and reading order. The native locality mechanism comes from [2], and the finite arbitrary-width slab geometry from [3]. The additional work is the paid replay across different original executions, the actual simulator pullback, and the carrier and reference constructions. We reuse standard counting and entropy inequalities. No external-priority claim is made for the combined statements.

For the direct native bounds, read Sections 3 to 5. For the every-output crossing and entry-capacity route, read Appendices G and H. Sections 6 and 7 give the simultaneous-store condition and its redundancy control. The other appendices develop coherent clusters, catalogues, local getters, reference changes, persistence and feedback. Section 8 specifies which additional statements would turn these requirements into an impossibility proof.

2 Native executions and records

We use the frozen native model of [2]. A machine has k two-way work tapes, alphabet Γ containing blank 0, and Q states. Thus $Q = \texttt{native.q} + 1$, not `native.q`. Put $a = \max(1, \lceil \log_2 |\Gamma| \rceil)$ and $q = \lceil \log_2 Q \rceil$. Every head starts at 0 and every work tape is blank. In a successful transition the machine reads the complete old tuple, writes at the old work-head positions, moves each head by $-1, 0$ or 1, and appends at most one bit. A halt test is not a successful transition.

We allow arbitrary integers $d, b \geq 1$ and write $P = d^2 b$. The canonical-width length is $L = d^2 b(d)$, where $b(d) = \max(1, \lceil \log_2 d \rceil)$. Let π send an output ordinal to its demanded raw coordinate; for transposition it exchanges the row and column indices and preserves the bit lane. Some replay statements hold for any fixed permutation π.

For an input X, write $x(t), y_h(t), w_h(t, z), s(t)$ for the input position, work positions, work symbols and state just before successful step $t < T$. Let $o(t)$ be the number of bits appended strictly before t. A return (p, t) of head h has $p + 2 \leq t < T$, $y_h(p) = y_h(t)$, and no intermediate visit to that cell. Its excursion is

$$\max_{p<u<t} |y_h(u) - y_h(p)|.$$

Adjacent repeated visits cannot be long returns.

The backward dependency graph has a state edge $t \to t - 1$ for $t > 0$ and a work edge from t to each actual last prior visit to its scanned work cell. A preserving write is still a visit and a version parent. Raw observations are attached to their read steps; there is no mutable input-tape edge.

Cuts delete whole vertices, including roots and targets. Write $\mathrm{Read}_X(U)$ for every successful raw read in U, and $\mathrm{Out}_X(V)$ for the actual append steps demanding coordinates in V. Paths of length zero count, so a root that is also a target must be deleted to separate it.

Lemma 2.1 (Old-position writes)**.** Lean: `step_work_update`, `step_scanned_write`, `step_preserves_unscanned`, `instruction_eq_of_old_tuple` [`NativeWrites.lean`]

After a successful native step, every unscanned cell is unchanged. At the scanned cell the new symbol is the instruction's write. Equal old state, raw symbol and scanned work tuple give the same instruction, regardless of unobserved tape contents and previously emitted output.

Outline of the formal proof. Unfold the frozen transition definition. It replaces the work symbol exactly when the queried address equals the old head position. The transition table does not receive the output history. □

All counts below concern one fixed transition table at each length. Uniform resource statements allow a nonuniform table only where explicitly stated; they never take a union over all tables. Correctness means the complete P-bit output, not merely agreement on a selected collection of outputs.

3 Replay without a common trace

Fix public integers $T_* \geq P$, $W \geq 1$ and $H \geq 0$. On a timely correct input choose a free head f, a slab $S = [\ell, \ell + W) \cap \mathbb{Z}$, and an outer interval $[a_0, b_0) \subseteq [0, T)$. Retain

$$C = \{t \in [a_0, b_0) : y_f(t) \in S\},$$

whose nonempty maximal runs are $[a_i, e_i)$, $1 \leq i \leq r$. The choices may depend on the input.

The geometry message records f, ℓ, a_0, b_0, r and, for each run, its time endpoints, all entry head positions, the raw-head minimum and maximum, and $o(a_i), o(e_i)$. Unit motion implies that the raw positions visited in a run are exactly the integer interval between its extrema. Consequently the message determines the raw coordinate set I read in C, the output ordinals O appended in C, and

$$U = \pi(O) \setminus I. \tag{1}$$

The detailed paths and append flags inside the runs are not recorded.

Lemma 3.1 (Incoming first reads)**.** Lean: `outside_first_read_is_long_return` [`IncomingReturns.lean`]

Suppose t is the first visit during $[a, e)$ to a cell outside the radius-H window about $y_h(a)$. If the cell has a previous global visit p, its last such visit satisfies $p < a < t$, and (p, t) is a return with excursion greater than H, witnessed by the entry time a.

Outline of the formal proof. A first run-visit cannot have its last predecessor inside the run. The entry time itself cannot be the read because it lies at the window center. These strict inequalities produce a genuine nonadjacent return; no $H + 1$ loss is needed. □

For every nonfree head, tag precisely the first reads in each run that are outside the entry window and have a previous global visit. Attach their old symbols to their typed slots (t, h). There are at most as many tags as actual off-carrier long returns, by Lemma 3.1; the actual typed-slot injection

is checked as `incoming_support_le_returns`. If at most J tags are used, one joint event/value menu has size

$$\mathcal{S}(n, J, A) = \sum_{j=0}^{\min(J,n)} \binom{n}{j} A^j, \qquad n = (k-1)T_*, \quad A = |\Gamma|. \tag{2}$$

One may enlarge n to kT_*. Repeated consumers of different versions are different events. There is no independent $J \log P$ address payment.

Lemma 3.2 (Joint sparse records)**.** Lean: `sizedImports_card`, `sparseImports_card`, `replay_message_bound` [`RecordCounts.lean`]

The finite type choosing at most J of n typed slots and one of A values for each chosen slot has the cardinality in Eq. (2). An injective encoding into this type, M metadata choices, e exterior bits and p payload bits bounds its input set by $M2^{e+p}\mathcal{S}(n, J, A)$.

Outline of the formal proof. For exactly j slots choose their subset and list the values in increasing slot order. The respective cardinalities are $\binom{n}{j}$ and A^j. Sum over j; terms with $j > n$ vanish. The second claim is the cardinality inequality for an injection into the explicit product type. Construction of the native encoding is proved next from the actual successful executions. □

Theorem 3.3 (Finite adaptive replay)**.** Lean: `native_fixed_geometry_replay_bound`, `integerSlab_card` [`ReplayGeometry.lean`]

For fixed geometry with r runs, let $\iota = 1$ if $a_0 > 0$ and 0 otherwise. The number of timely correct inputs represented by that geometry and at most J tagged imports is at most

$$2^{P-|U|} Q^r A^{\iota W + (k-1)r(2H+1)} \mathcal{S}((k-1)T_*, J, A). \tag{3}$$

This statement permits different original traces within the same message class.

Outline of the formal proof. The encoder records the input outside U, each run-entry state, the nonfree radius-H entry windows, the free slab at $a_0 > 0$, and the actual joint typed import record. A tagged blank is a present value, distinct from an absent tag. At $a_0 = 0$ the slab record has an empty domain.

Compare two actual successful executions with equal records. A simultaneous induction proves equality of their retained control and instructions, and of every free-slab cell throughout the outer interval. An omitted step writes outside the slab in both executions. At a retained step, a nonfree read agrees by its last in-run write, its paid entry window, or its equal incoming tag. For a first outside-window read with no tag, the actual cell is virgin and blank. The raw symbols agree by the exterior record and the recorded extrema, with public endmarkers outside the input. Equality of the complete old tuple determines the simultaneous native transition. These are the checked declarations `ExecutionPrefix.adaptive_replay_agreement` and `ExecutionPrefix.replay_output_count_agreement`.

The maximal-run decomposition supplies the entry map and every retained prefix. The native output counter increases by zero or one, so every ordinal between a run's recorded entry and exit counters has an actual append in that run. This derives output coverage from the geometry itself. Equal instructions and entry counters then give equal retained append ordinals and bits. Actual final-output correctness recovers U; the exterior gives the other coordinates. The resulting `encodeReplay_injective` compares actual runs directly and assumes no common detailed trace or decoder.

The state, window and slab products have exactly the displayed sizes. `replayImports_card_le` bounds the actual Option-valued sparse tuple by the joint support/value menu, including its event

positions. Terms above the number of typed slots vanish. The formal theorem allows any finite free cell set and any demand map π; a width-W integer slab has exactly W cells and gives Eq. (3). □

If $r \le N$, the following safe metadata menu will be used:

$$\mathcal{M} = k(N+1)(2T_* + W + 2)(T_* + 1)^2\big[(T_* + 1)^4(2T_* + 1)^{k+3}\big]^N. \tag{4}$$

A nonempty slab has $-T_* - W + 1 \le \ell \le T_*$. Every head coordinate and raw extremum lies in $[-T_*, T_*]$; every time and output counter is in $[0, T_*]$. There are two outer endpoints, and per run two time endpoints, two output counters, $k+1$ entry positions and two extrema. Padding to N runs gives Eq. (4). The actual bounded encoder and exact cardinality are checked as `SlabObservation.encode_times` and `replayMetadata_card`. All correlations only reduce this product bound. At fixed k, N and $T_*, W = O(P)$ its logarithm is $O(\log(P+1))$.

4 Macroscopic returns on original adaptive runs

Fix $k, a \ge 1$ and $C \ge 1$, and put

$$T_* = \lfloor C(P+1) \rfloor, \qquad W = \lceil P/(80aC) \rceil, \qquad N = \lfloor 160aC^2 \rfloor + 1.$$

Let $\mathrm{Bad}(H, J)$ consist of timely correct inputs for which some free head leaves at most J returns of excursion greater than H on the other heads.

Lemma 4.1 (Finite geometry)**.** Lean: `exists_slab_geometry`, `native_slab_geometry` [`SlabGeometry.lean`]

For $P \ge 2$, every timely correct run and chosen free head has a slab chunk C of at most $\lfloor P/2 \rfloor$ steps and at most N runs. With U from Eq. (1), it satisfies

$$|U| - a\iota W \ge P/(4N) - 3Ndb - a. \tag{5}$$

Outline of the formal proof. We use the arbitrary-width slab and stripe lemmas of [3], Lean-verified there as

- `NativeRelations.exists_offset_slabTransitions_le`;
- `NativeRelations.exists_slab_chunk_dichotomy`;
- `NativeRelations.outIn_card_le_two_indep_add_six_runs`.

The actual-input adapter obtains the halting time and all P append events from timely transposition of that input. Its own raw and work paths are unit walks, and its actual append times are strictly increasing. No other input is required to share this trace. The finite theorem constructs literal slab intervals from these walks and times; it does not assume a suitable chunk. Their finite statements imply this particular margin as follows. An averaged grid offset has at most $(T-1)/W < N$ wall crossings. The short case takes all visits to its slab from time zero, has at least $P/(2N)$ outputs and pays no snapshot. The long case takes a time slice and has $P\lfloor P/2 \rfloor < 4T|O|$. In either case the stripe estimate is $|O| \le 2|U| + 6Ndb$. Since $P \ge 2$, $T \le 2CP$ and $\lfloor P/2 \rfloor \ge P/3$, the long case gives

$$|U| - aW > P/(48C) - P/(80C) - 3Ndb - a = P/(120C) - 3Ndb - a.$$

Now $N > 160aC^2 \ge 160C$ implies Eq. (5). The short and long cases must be distinguished: charging a snapshot against the short-case output bound would not justify this margin. □

Theorem 4.2 (Finite original-run bound)**.** Lean: `native_positive_saving_observation`, `native_finite_return_bound` [`NativeReturnCounting.lean`]

For $P \geq 2$, using Eq. (4) and $A \leq 2^a$,

$$|\operatorname{Bad}(H,J)| \leq 2^{P-P/(4N)+E}, \qquad E = 3Ndb + a + a(k-1)N(2H+1) + Nq + \log_2 \mathcal{M} + \log_2 \mathcal{S}(kT_*, J, A). \tag{6}$$

There is no all-input correctness or common-trace assumption.

Outline of the formal proof. When the snapshot-adjusted margin is positive, the native slab theorem selects a nonempty slab observation for each input and qualifying free head. The encoder constructs its bounded metadata from the actual maximal runs, head positions, raw extrema and append counters. Equal metadata determines the retained time set and all these observations, while allowing different detailed native traces. Unit-path continuity identifies the raw-extrema union with the actually read cells; the output-counter intervals identify exactly the actual appends.

The checked `constantSlabClass` constructs the replay class on every metadata fibre. Its successful prefixes and correct final words come from the inputs' own native runs. The actual typed import support injects into completed off-carrier long returns, so at most J imports are paid. The snapshot-adjusted saving is Eq. (5); the remaining windows and states cost at most $a(k-1)N(2H+1) + Nq$. Sum the class bounds over the explicit menu of cardinality $\mathcal{M}$. Each finite menu contributes one logarithm. If the geometric margin is nonpositive, the full 2^P input bound already implies the claimed inequality. The formal endpoint applies to any set of timely correct inputs satisfying the stated return condition, without all-input correctness or a common detailed trace. □

Theorem 4.3 (Theorem R: two macroscopic return heads)**.** Lean: `returnTailEpsilon_pos`, `returnTailEta_pos`, `native_macroscopic_return_tail` [`ReturnTail.lean`]

For fixed k, a, C there are $\varepsilon, \gamma, \eta > 0$ and a dimension threshold uniform in all $b \geq 1$ such that fewer than $2^{(1-\eta)P}$ timely correct inputs admit a free head with at most γP off-carrier returns of excursion greater than $\lfloor \varepsilon P \rfloor$. This holds for each individual machine with $q \leq P/(64N^2)$. In particular, for $k \geq 2$, every other timely correct input has two distinct heads each making more than $\gamma P/(k-1)$ such returns. For $k = 1$ the bound applies to the entire timely success set.

Outline of the formal proof. Take $\varepsilon = [64a \max(1, k-1)N^2]^{-1}$ and $\eta = 1/(8N)$. The checked sparse-record estimate `native_sparse_logb_small`, applied to the alphabet bound 2^a, gives one $\delta > 0$ such that $J \leq \delta P$ implies $\log_2 \mathcal{S}(kT_*, J, A) \leq P/(32N)$. Choose $\gamma = \min(\delta, 1/2)$ and $J = \lfloor \gamma P \rfloor$. This includes $J = 0$. The floor bound and the stated ε give

$$a(k-1)N(2\lfloor \varepsilon P \rfloor + 1) \leq P/(32N) + a(k-1)N.$$

The actual metadata menu satisfies the checked polynomial bound

$$\mathcal{M} \leq k(N+1)\big[(2\lceil C \rceil + 3)(P+1)\big]^{3+(k+7)N}.$$

Thus `native_metadata_logb_small` supplies a length threshold P_0 with $\log_2 \mathcal{M} \leq P/(192N)$ for $P \geq P_0$. One valid dimension threshold is

$$d_0 = 2 + 576N^2 + P_0 + 192N\big[a + a(k-1)N\big].$$

For $d \geq d_0$ and every $b \geq 1$, $P = d^2 b \geq d$; the three terms $3Ndb$, $a + a(k-1)N$ and $\log_2 \mathcal{M}$ are each at most $P/(192N)$. Their sum is at most $P/(64N)$. The controller budget contributes another

$P/(64N)$, and the proportional window and sparse costs each contribute $P/(32N)$. The exact finite bound therefore has exponent at most $(1-5/(32N))P < (1-\eta)P$. Every threshold is independent of b and of the individual machine within the stated resource budgets.

For a good input, deleting any one head leaves more than γP returns; the strict inequality follows from the integer floor cutoff. Partition the actual return triples by head and choose a head of maximum count. Among the other $k-1$ heads one has more than $\gamma P/(k-1)$ returns, and the maximum head has at least that many. These are the checked declarations `offCarrierReturns_card_eq_sum`, `native_two_return_heads` and `native_one_head_bad`. In the one-head case the off-carrier set is empty, so every timely correct input belongs to the bad set. □

Fixed controllers, and uniformly $o(P)$-bit controllers, eventually meet the displayed controller budget.

5 Causal fragments and original cuts

Lemma 5.1 (Fragment starts)**.** Lean: `fragment_count_le`, `localized_fragment_count_le` [`Fragments.lean`]

If a finite retained set K is closed under state predecessors except at a cut B, it has at most $|B|+1$ nonempty runs. If closure is required only inside a selected set C and $K \subseteq C$, it has at most $|B|+r(C)$ runs, where $r(C)$ is the number of runs of C.

Outline of the formal proof. A noninitial run start must immediately follow a cut. In the localized case it may instead be a run start of C. Map the other starts injectively to their preceding cut times. □

Lemma 5.2 (Current versions suffice)**.** Lean: `filtered_read_correct`, `ExecutionPrefix.filtered_native_read` [`CurrentVersion.lean`]

Process an actual native execution's chosen writes in chronological order, starting with blanks. At a requested read the resulting dictionary gives the actual symbol if the actual last prior visit to that cell is processed, or if no prior visit exists. No equality on other cells is required.

Outline of the formal proof. The recursive dictionary is unchanged on an interval with no write to the queried cell. If a last visit exists, both the full and filtered dictionaries therefore equal its written value. Otherwise both equal the initial value. The native statement follows by induction from Lemma 2.1. □

Let $\beta = q + ka + 2$ encode the complete step value: poststate, all work writes and optional output bit. Old positions and move trits are paid separately. For $s \geq 0$ put $n = s+1$ and

$$\mathcal{J}_s = (T_*+1)\binom{T_*}{s}\binom{T_*+n}{n}\left[6^s\binom{T_*+s}{s}\right]^{k+1}\binom{T_*+2n}{2n}\binom{P+4s+2}{4s+2}. \tag{7}$$

The exact menu in Eq. (7) is checked by `globalGeometryMenu_card`. The declarations `sampledWalk` and `headDescription_injective` construct the signed-increment description from chronological samples of a unit walk. Monotone counters are encoded by their multiset, whose sorted order recovers every sample. The actual native encoder and its raw-coordinate and counter recovery are checked in `FragmentGeometry.lean`; `totalFragmentRecord_card` in `FragmentMenuCount.lean` combines them with the literal cut set and complete cut payload.

Theorem 5.3 (Finite adaptive input-cut bound)**.** Lean: `mem_nativeBadInputCuts_iff`, `native_bad_input_cut_bound` [`NativeInputCutEndpoint.lean`]

For any fixed output permutation π, let $\mathrm{BadInput}(D,r)$ *be the timely correct inputs admitting sets $V \subseteq U$ with $|V| \geq D$ and a vertex cut B, $|B| \leq r$, from* $\mathrm{Out}_X(V)$ *to all of* $\mathrm{Read}_X(U)$*. Then*

$$|\,\mathrm{BadInput}(D,r)| \leq 2^{P-D} \sum_{s=0}^{\min(r,T_*)} \mathcal{J}_s 2^{\beta s}. \tag{8}$$

The sets and cut may depend on the input and are not separately encoded.

Outline of the formal proof. Retain the ancestors K of the noncut desired roots in the graph with B deleted. In each B-free gap the state spine makes K a prefix. Every work parent of a retained step belongs to $K \cup B$, unless the cell is virgin. Record every cut value, old position and move, and the retained prefix lengths. At a retained interval entry the state and heads are initial, or are obtained from the immediately preceding cut step. Process retained transitions and recorded cut writes in time order. By Lemma 5.2, an omitted overwrite cannot remain the latest parent of a retained read; closure would have retained or cut it. A simultaneous induction compares two actual executions with the same record. Their retained instructions agree because their old tuples agree; no common detailed trace is assumed.

Record raw extrema for each retained interval and the before/after output counters for every retained interval and every cut step. Let I be the raw coordinates read in K, and O *all* output ordinals at $K \cup B$. The paid record determines both, including append flags at cut steps through their two counters. Record the exterior of $W = \pi(O) \setminus I$. Retained raw reads are known. Cut outputs are paid directly. Replay recovers the outputs O and therefore W, hence the entire input. The separator gives $I \cap U = \varnothing$ and $V \subseteq \pi(O)$, so $V \subseteq W$. Full correctness at all of O is essential to this description-free step. The formal event is expanded into an actual first halt, the complete output, and the literal endpoint-inclusive separator by `mem_nativeBadInputCuts_iff`.

Here is the exact menu count. The horizon and cut set cost $(T_*+1)\binom{T_*}{s}$. The n prefix lengths have total at most T_*, giving $\binom{T_*+n}{n}$. For each head, successive sampled cut positions have total absolute variation at most T_*, giving $2^s\binom{T_*+s}{s}$ signed-increment descriptions; moves add 3^s. The $2n$ nonnegative raw-extremum offsets have total at most T_*, giving $\binom{T_*+2n}{2n}$. Finally the $4s+2$ chronological output-counter samples are weakly increasing in $[0,P]$, giving $\binom{P+4s+2}{4s+2}$. Empty intervals and invalid descriptions only overcount. These factors are Eq. (7). The cut payload costs βs bits, and the exterior at most $P-D$ bits. Replay proves the injection separately for every actual execution sharing the message; it does not fix one input's query set for all other inputs. Sum over s. □

Corollary 5.4 (Theorem I: simultaneous source-input expansion)**.** Lean: `native_input_cut_tail`, `native_input_cut_tail_all_width` [`InputCutTail.lean`]

Fix k, β, a runtime constant $C \geq 0$, and $\alpha > 0$. There is $\lambda > 0$ such that at all sufficiently large lengths fewer than $2^{(1-\alpha/3)P}$ timely correct inputs admit any cut in Theorem 5.3 with $|V| \geq \alpha P$ and $|B| \leq \lambda P$. For transposition the threshold is uniform over all $b \geq 1$ as $d \to \infty$.

Outline of the formal proof. Put $r = \lfloor \lambda P \rfloor$. Bound every binomial factor in the exact menu by the common sparse sum with $4r+2$ chosen positions out of $P + T_* + 4r + 2$. The full cut menu is at most

$$(r+1)(T_*+1)\left[6^{k+1}2^{\beta}\right]^r \left[\sum_{j=0}^{4r+2} \binom{P+T_*+4r+2}{j}\right]^{k+5}.$$

The generating-function sparse estimate and the polynomial logarithm bound make its base-two logarithm at most $\alpha P/2$ after choosing one positive λ and then a length threshold. This is checked by `geometry_cut_menus_logb_small` in `GeometryMenuRates.lean`; it requires no $r \le T_*/2$ restriction. The finite native bound gives $2^{(1-\alpha/2)P} < 2^{(1-\alpha/3)P}$ for $P > 0$. The proof uses $T_* \le \max(1, \lceil C \rceil)(P+1)$ and $P = d^2 b \ge d^2$, so one threshold works for every positive block width. No union over U, V is needed. □

5.1 Localizing the same decoder

For a selected slab chunk C from Lemma 4.1, keep its geometry and ordinary entry records. Put $U = \pi(O(C)) \setminus I(C)$. Let Y_H be *all earlier endpoints* of off-carrier returns of excursion greater than H, including endpoints outside C. Suppose B separates $\mathrm{Out}(U)$ from Y_H in the full original graph. Retain the ancestors K of the noncut roots in the graph induced on $C \setminus B$. There are at most $N + |B|$ retained fragments. A parent in C is in $K \cup B$; any parent in Y_H is in B, even when it is outside C, by the full separator property.

Process ordinary snapshots as pre-step overlays, and retained/cut writes as step events. At the outer start install the free-slab snapshot or blanks. At each original C-run entry overlay the actual nonfree windows. Do not erase dictionary cells outside these windows. At ties perform overlays before the cut or retained write. Fragment entry states and heads come from an original run entry or the preceding cut step; record their output counters separately.

For a retained read on the free tape, its last visit after the outer start belongs to C, hence to $K \cup B$; an older version is covered by the initial slab snapshot. On a nonfree tape, a last visit in the current C-run is in $K \cup B$. An earlier last visit is covered by the entry window, or lies in Y_H by Lemma 3.1 and is therefore in B. A virgin cell is blank. These cases supply the correct current version. Later true snapshots cannot spoil it: after the last visit and before this read every actual snapshot of that cell has the same value. This also handles cut steps before the outer start, cut roots and snapshot/cut ties. It would be false to claim that a whole core dictionary remains correct after skipping $C \setminus K$.

The raw exterior is determined by the original C, so no new extrema for the smaller fragments are required. With $s = |B|$, their extra geometry has the safe menu

$$\mathcal{L}_s = (T_*+1)\binom{T_*}{s}\binom{T_*+N+s}{N+s}\left[6^s\binom{T_*+s}{s}\right]^{k+1}\binom{P+4s+2N}{4s+2N}. \tag{9}$$

Its exact finite product cardinality is `localGeometryMenu_card`. The actual native local record and its recovery are checked in `LocalGeometry.lean`. The bound `native_bad_slab_cut_bound` combines the derived native injection with the original slab metadata. The factors count cut times, the $N+s$ prefix lengths, sampled positions and moves, and $2(N+s)+2s$ output-counter samples. The proof is the same joint sparse count as above. Replacing the import menu in Eq. (6) by $\sum_{s \le r} \mathcal{L}_s 2^{\beta s}$ therefore bounds the inputs with such a cut of size at most r.

Theorem 5.5 (Theorem H: original off-carrier expansion)**.** `Lean`: `native_off_carrier_expansion` [`OffCarrierExpansion.lean`]

For fixed k, a, q, C there are $\varepsilon, \lambda, \eta > 0$ and a threshold uniform over all $b \ge 1$ such that, outside fewer than $2^{(1-\eta)P}$ timely correct inputs, every choice of free head admits a demanded raw set U with $P/(8N) \le |U| \le P/2$ whose original minimum vertex cut to all off-carrier $\lfloor \varepsilon P \rfloor$-long return lower endpoints is greater than λP. In fact every valid chunk of Lemma 4.1 has this property on the same good-input event.

Outline of the formal proof. The exhaustive current-version induction above gives the actual native pairwise injection in `NativeReplayClass.LocalCut.records_recover` of `NativeLocalCuts.lean`. Its last-parent and desired-output coverage are derived from the original full separator and actual counters. Inputs sharing computed slab metadata have the same omitted set, even when they halt at different later times. Group those actual inputs into metadata fibres, pay the ordinary snapshots once, and sum their local-record bounds. No common detailed trace or import bound is imposed.

Take $H = \lfloor \varepsilon P \rfloor$ with the explicit ε of Theorem 4.3. The common sparse envelope from `geometry_cut_menus_logb_small` makes the exact local cut menu's logarithm at most $P/(32N)$ after choosing one positive λ and a length threshold. The windows cost at most $P/(32N)$ apart from fixed terms. The metadata, the $3Ndb$ term, and all fixed terms including Nq each cost at most $P/(96N)$ beyond a single dimension threshold. The margin $P/(4N)$ therefore leaves exponent at most $1 - 5/(32N)$, strictly below $1 - \eta$ for $\eta = 1/(8N)$. This is `native_local_cut_tail` in `LocalCutTail.lean`.

The one bad event ranges over every valid actual chunk, so it already includes all free-head and chunk choices. At the enlarged threshold the geometric saving is at least $P/(8N) > 0$; the proved slab-selection theorem therefore supplies a nonempty valid chunk for every specified free head. Its margin gives $|U| \geq P/(8N)$, and injectivity of actual append times gives $|U| \leq |O(C)| \leq |C| \leq P/2$. Decreasing H only adds original targets, so the same U works for every smaller nonnegative integer threshold. All length thresholds use $P = d^2 b \geq d$, uniformly for $b \geq 1$. □

The cut-density constant depends on the whole value width β, hence on the controller. Unlike Theorem 4.3, this fixed positive cut-density consequence does not automatically extend to arbitrary $o(P)$-bit controllers. No result here supplies a small separator for every linear-time execution.

6 A dynamically changing sparse carrier

The previous theorems constrain motion and communication. We next obtain a simultaneous storage condition with no assumption that the exceptional tape has a fixed identity.

Theorem 6.1 (Exact dynamic-carrier simulation)**.** Not formalized; a complete proof is given below. *Fix a native machine with $k \geq 1$ tapes, Q states and alphabet $A \leq 2^a$. Fix integers $T, m \geq 0$, put $n = 2T + 1$, and define*

$$V = \sum_{j=1}^{\min(n,2m+2)} \binom{n}{j} (2^a - 1)^j, \qquad r = \lceil \log_2(V + 1) \rceil,$$
$$B_0 = k(T+1)n^{k+1}(n+1)^k(4T+1)(1+n^2).$$

There is one native machine with one work tape and the same alphabet that, on every source input halting within T for which every reached configuration has at most one tape with more than m nonblank cells, reproduces exactly the source's successful-transition count, raw-head moves and append times/bits. It has at most $1 + QB_0 2^{(k+1)r}$ states. It may fail or halt off the promise. The table depends on the source machine, T, m, not on the individual input.

Proof (not formalized). Embed the alphabet in $\mathbb{F}_2^a$, with blank zero. A tape supported in $[-T, T]$ is a vector with n blocks. There are at most V nonzero vectors of block weight at most $2m + 2$. A uniformly chosen binary $r \times an$ matrix kills each with probability 2^{-r}, so some matrix H kills none. Fix it. Its syndrome is injective on legal tape words of weight at most $m + 1$. The partial inverse is a finite transition-table function; it does not assert that a dense tape can be decoded. This

finite selection is checked in `binary_matrix_collision_card` and `exists_sparse_injective_hash`; `binary_hash_bits_eq_ceiling` identifies the bit-length convention. The remaining native transition table and handoff invariant below are not yet formalized.

Maintain all k full-tape syndromes and exact nonblank counts in control. Replacing old symbol u by v at address z updates the syndrome by $H(e_z \otimes (\mathrm{enc}(u) + \mathrm{enc}(v)))$ and the count by $1_{v\neq 0} - 1_{u\neq 0}$. The syndrome identity is the specialization of `LeanMachine.linearHash_update` to the alphabet embedding. Store all logical positions, the source state and time, an active tape f, physical position p, offset u_0 with $p + u_0 = y_f$, a frozen phase-entry syndrome, and an initially empty interval I of physical cells previously scanned in this phase.

The invariant before a source step is: every inactive tape is m-sparse; the physical symbol at $j \in I$ is the current logical symbol of f at $j + u_0$; outside I the logical symbol is given by the frozen phase-entry word, whose weight is at most $m + 1$. The syndrome registers always encode the current whole logical tapes, including a dense active tape. Initially all words and registers are zero, $f = 0$, $u_0 = 0$ and $I = \varnothing$.

Read the old inactive symbols by sparse decoding. Read the active symbol physically if $p \in I$; otherwise ignore the stale physical symbol and use the frozen snapshot. This gives the entire original old tuple. Apply the source transition once. If it halts, halt without a successful step. Otherwise update every logical syndrome, count and position. Physically write the old active tape's prescribed new symbol at p and take its prescribed unit move. Take the original raw move and optional append in this same transition.

If the updated configuration has no dense tape, or its dense tape is still f, retain the carrier, offset and snapshot, and replace I by $I \cup \{p\}$. Unit motion makes this a single interval; its new cell was just written to the correct value. If a different tape g has become the unique dense tape, it was inactive and m-sparse before the transition. One write raises its count by at most one, so its new count is exactly $m + 1$. Decode its updated syndrome as the new frozen snapshot, choose the new offset $u_0' = y_g' - p'$, and reset I to empty. The old active tape is now sparse and can be decoded from its updated live syndrome. The physical move was still the old carrier's move; the new offset requires no second move, seek, clearing scan or copy. Stale physical cells are ignored until first visited and overwritten. This proves the invariant through arbitrary handoffs, including simultaneous changes on both involved tapes.

All choices are finite table operations. Illegal decodes, multiple dense updated counters and out-of-range fields may halt; promised executions never take these branches. Force a halt when the stored time is T. A promised source already halts there or earlier. The physical and logical positions are bounded by T, the offset by $2T$, and I has at most $1 + n^2$ possible empty/endpoint descriptions. Time, carrier, positions, counts and offset have exactly the factors overcounted in B_0. There are k live syndromes and one frozen syndrome. Multiplying by Q and adding a default state gives the stated bound. This is an actual finite native table with one physical work head; it is nonuniform in T, m, and no small table-description claim is needed. □

Theorem 6.2 (Theorem S: simultaneous large stores)**.** Lean (partial): `stores_tail_of_carrier`, `native_two_dense_heads`, `native_support_window_complete` [`StoresFromCarrier.lean`]
Not formalized: The exact state-bit budget and success tail are checked for one common native carrier simulator with the stated state count and halt/output correspondence. Constructing that simulator, as in Theorem 6.1, remains unformalized.

Fix k, a, C. There are $\delta, \eta > 0$ and d_0, uniform over all $b \geq 1$, such that fewer than $2^{(1-\eta)P}$ timely correct inputs fail to reach a configuration with two distinct tapes each containing more than δP nonblank cells. This holds for fixed controllers and, more generally, for $q \leq P/(128N_1^2)$, where $N_1 = \lfloor 160aC^2 \rfloor + 1$.

Proof (not formalized). Apply Theorem 6.1 to precisely the successful inputs satisfying the one-dense-tape promise, with $T = \lfloor C(P+1) \rfloor$ and $m = \lfloor \delta P \rfloor$. The one-head case of Theorem 4.3 applies to each nonuniform simulator if its state-bit cost is at most ρP, $\rho = 1/(64N_1^2)$. Write $K = 4C + 1$. At large P, $2m + 2 \le 4\delta P \le n/2$ and

$$r \le 1 + 4\delta P \log_2 \frac{eK(2^a - 1)}{4\delta}.$$

Choose $0 < \delta \le 1/8$ so that the last coefficient multiplied by $k + 1$ is at most $\rho/4$. The logarithm of B_0 is $O(\log(P + 1))$. The source state cost is at most $\rho P/2$, leaving sufficient room for this overhead and ceilings. The checked `carrier_state_bits_small` proves the complete integer bit budget directly from the sparse-menu generating-function bound, including the default state and every syndrome. Thus the simulator satisfies the one-head budget and has fewer than $2^{(1-1/(8N_1))P}$ timely correct inputs. Every promised successful source input is among them, with no loss and no per-input simulator choice. All thresholds depend only on the stated resources, because $P \ge d^2$. The checked complement lemma converts the floor cutoff to the strict real threshold at one common reached configuration, including the halt configuration. The finite window $[-T, T]$ contains the entire native nonblank support. □

Lemma 6.3 (Persistent unchanged cells)**.** Lean: `native_support_growth_le_one`, `native_support_shrink_le_one`, `ExecutionPrefix.untouched_cell_constant`, `ExecutionPrefix.persistent_cells` [`Storage.lean`]

For any fixed finite window on a tape, one native transition changes its nonblank-cell count by at most one. In an interval of ℓ successful steps, at most ℓ terminal nonblank cell addresses in that window were scanned in the interval; every other terminal nonblank cell in the window keeps its symbol throughout the interval.

Outline of the formal proof. Outside the old scanned address, Lemma 2.1 preserves the symbol. The touched set is the image of the interval's times under the head-position map and has cardinality at most its duration. Remove it from the terminal support. The formal statement uses any finite window containing the cells; all live cells are in $[-T, T]$ on a T-step run. □

At the witness time of Theorem 6.2, both counts exceed δP. They started at zero, so the time exceeds δP. Throughout the preceding $\lfloor \delta P/2 \rfloor$ steps both counts exceed $\delta P/2$. Deleting the addresses written in that interval leaves more than $\delta P/2$ fixed nonblank cells on each tape. Their values need not be independent, unknown, necessary for output, or even read again. The checked `ExecutionPrefix.dense_persistent_interval` gives the more general exact loss of ℓ cells over a preceding interval of ℓ steps.

7 Construction side: redundant archives

The necessary conditions above do not imply that two tapes contain independent information. The following paid normalization rules out that interpretation, even when the selected words are future-readable and far from the heads.

Theorem 7.1 (Redundant-store normalization)**.** Not formalized; a complete proof is given below.

For every fixed native machine M with $k \ge 1$ work tapes there is a blank-started native machine $\widehat{M}$ with the same tape count and fixed enlarged alphabet/control that has the same output on every halting nonempty input and halts there within $8T_M(X) + 20|X| + 40$ steps. It halts iff M does. Fix an input length P, any probability law on its halting inputs, and any absolute configuration time,

padding after halt. For any two tapes, their whole words mutually determine one another. The same is true of their future-readable words, and of their future-readable words with a common closed radius-H head window blanked, whenever $2H \leq P$. Head positions are not included in these word variables.

Proof (not formalized). Use fixed product tracks for the original payload, tagged raw archives, an origin marker, an end marker and a visited flag. Tagged zero differs from blank. All heads prepare synchronously. Copy the raw input into tagged cells $[0, P-1]$ on every tape, marking the origin at zero. At the right raw marker move raw left and work heads right, then continue while raw scans bits. A turn moves raw from -1 to zero while work heads stay at $2P+1$. Copy the input again into $[2P+1, 3P]$. Mark the end at $3P+1$, then return all work heads to the origin and independently return raw to zero. Counting the bit, turn and phase transitions gives $6P+4$ preparation steps. There is no length counter, extra metadata head or preloaded tape. Payload is still blank.

Simulate each original transition on its complete old payload tuple in one step, preserving the archival tracks. Maintain one optional pending output bit: on an original append, emit the old pending bit if present and replace it with the new bit. Otherwise emit nothing. The emitted word followed by the pending bit equals the original output. This uses three pending states and never reads output. Mark every old scanned cell as visited, including each preparation/core step and the stationary wrapper transition detecting the original halt. Seen cells on each tape form one interval $[l_h, r_h]$ containing $[0, 3P+1]$, and every nonblank payload cell is inside it.

At halt, independently on each head, sweep left through visited cells while blanking payload but retaining the visited flag. At the first unvisited guard turn right. Sweep right through the whole visited interval, blanking payload and clearing its visited flag behind the head. At the right unvisited guard turn left and return to the unique origin. Park completed heads. These are finite local modes; all heads work in parallel. Archives and markers are preserved. The seen interval has left endpoint at least $-T_M(X)$ and right endpoint at most $\max(3P+1, T_M(X))$. A safe joint cleanup cost is $5T_M(X) + 6P + 10$. At its end every tape has exactly the same word: the two archives, public markers and blanks elsewhere. A final synchronized scan from origin through end successfully reads both archives, then appends the pending bit and halts. Together these costs are below the stated bound.

Before the public end of preparation, all tape words and heads are identical. Every then-nonblank cell is an archive or public marker visited by the final successful scan, so the future-readable words are identical as well. During the unsynchronized core and cleanup, both tagged archives remain in each future-readable word because of the final scan. A window of diameter $2H \leq P$ cannot meet both archives, whose nearest cells are distance $P+2$ apart. Thus each whole, future-readable or remote word contains an intact copy of X. During the synchronized tail and after halt, all same-type words are equal across tapes, since both contents and remaining trajectories agree.

This is also valid for a law mixing inputs in different phases at one absolute time. A same-type word containing an intact tagged copy determines X, hence the deterministic run and the desired other-tape word at that time. If it contains no intact copy, it cannot be in the unsynchronized core/cleanup, so return that same word: the tapes are then synchronized. The public preparation case is equality as well. These are mathematical decoders, not uncharged native access routines. Mutual deterministic dependence follows for every common input law. Divergent output prefixes need not be identical because a final bit can remain pending; only halting partial-function preservation and equivalent halting are asserted. □

On every timely transpose-correct input the source has at least P successful steps. During the common first P simulated steps each archived word is an injective function of the input, yet its

conditional information given any other archived word is zero. The output is still incomplete because of the pending-bit delay. Ignored future rereads can therefore satisfy large live marginal-information requirements. Excluding every such normalized transposer would already settle the fixed-tape-count existence question itself.

An affirmative construction must deliver the required output order through actual local reads while meeting the return, cut and storage conditions. Neither large joint support nor a globally available mathematical decoder pays those accesses. Conversely, an impossibility proof needs a physical supplier of a small cut, a cheap joint record or another output-sensitive deficit. None is supplied for unrestricted coupled mutable tapes here.

8 What remains for transposition

The question (T) remains open. A hypothetical linear-time transposer must meet the return, cut and storage requirements simultaneously. The direct bounds apply to its original adaptive execution at every positive width. The simulator pullback adds every-large-output crossing expansion and entry-capacity certificates at canonical widths. Thus these are constraints on the actual computation, not only on an auxiliary trace.

Two concrete routes remain. First, prove an output-specific upper bound on a cut or a jointly replayable boundary record that contradicts the corresponding lower bound. The record must retain current-version dependencies and raw reads, paying shared information once. Second, find a low-capacity certificate on the actual original entry relation. With the fixed s, δ of Corollary H.6, that certificate would have to give, for some B, J,

$$\Delta_{b-6s}(Q_X^B[J]) > \delta P + v_s G_s |B|.$$

The theorem gives the reverse inequality on every timely correct input outside its ambient exception. An exclusion must establish the conflicting certificate under the proposed class's actual execution hypotheses.

The decoder program of [1, Problem 9.4] is a possible source of such certificates, but it still needs an interface theorem. Our entry quotient does not automatically have shared decoders or a linear correctness source. A large arbitrary fibre does not supply those properties. Conversely, a construction must deliver all intermediate representations and the ordered transpose through paid native accesses, rather than merely exhibit a high-rate abstract code.

The other results distinguish useful premises from insufficient statistics. Coherent head clusters cannot hide all long-range communication; fixed catalogues and locally readable references cannot explain all but one store throughout the run. However, persistent marginal information may remain redundant across tapes, and cheap copying may satisfy the historical-feedback requirement. These controls rule out interpreting each large store or each reused version as an independent time charge. A superlinear bound still requires a justified way to combine obligations across the computation; no such general multiscale charge is proved here.

The remaining formalization obligations are separate from (T): the new carrier and catalogue compilers, raw-relative and historical extensions, last-visit eraser, redundant archive and block-monitor/spool construction. Their precise boundaries are recorded in Appendix J. The finite hash, menu, entropy and asymptotic steps, concrete scan loader, and simulator/source correspondences are checked.

A Coherent clusters of work heads

Several physical heads can share the protected-slab argument. Let $\varnothing \neq F \subseteq \{1, \dots, k\}$, $m = |F|$, choose fixed signs $\sigma_h \in \{-1, 1\}$, and put $u_h(t) = \sigma_h y_h(t)$ and $z(t) = \sum_{h \in F} u_h(t)$. A normalized chart consists of nondecreasing 1-Lipschitz maps $f_h : \mathbb{R} \to \mathbb{R}$ with $\sum_h f_h(v) = v$ for every real v. Its error is at most J if $|u_h(t) - f_h(z(t))| \leq J$ at every live step. The cluster, signs and maps may depend on the input, but remain fixed through that execution.

Lemma A.1 (Protected cores and paid fringes)**.** Lean: `selected_in_core_or_fringe`, `outside_avoids_core`, `protected_core_disjoint_fringe`, `normalized_chart_core_fringe_counts` [`ProtectedCores.lean`]

For $J \geq 0$, let $j = \lceil J \rceil$ and select a carrier slab $[\ell, \ell + W)$, and put $A_h = f_h(\ell)$, $B_h = f_h(\ell + W)$. On oriented tape h define

$$C_h = (A_h + j, B_h - j) \cap \mathbb{Z}, \qquad F_h = ([A_h - j, A_h + j] \cup [B_h - j, B_h + j]) \cap \mathbb{Z}.$$

Every selected read lies in $C_h \cup F_h$; every outside-slab step avoids C_h; $C_h \cap F_h = \varnothing$; and $\sum_h |C_h| \leq W + m$, $\sum_h |F_h| \leq m(4j + 2)$. Only four rounded endpoints per head are needed to encode these sets.

Outline of the formal proof. Inside the carrier slab monotonicity places $f_h(z)$ between A_h and B_h, and the error places the physical position in $[A_h - j, B_h + j]$. Its complement to the strict core lies in the two closed fringes, including when they overlap or the core is empty. If $z < \ell$, the position is at most $A_h + j$; if $z \geq \ell + W$, it is at least $B_h - j$. Thus no outside old position writes a core. A real interval of length $B_h - A_h$ contains at most that length plus one integer. Sum and use $\sum_h (B_h - A_h) = W$. Each closed fringe interval contains at most $2j + 1$ integers. Rounding by floor and ceiling determines all endpoints exactly; reflection by σ_h preserves the counts and ownership. Equality at a core boundary is deliberately assigned to the fringe. The checked bound also allows individual errors j_h, with total fringe bill $\sum_h (4j_h + 2)$. □

Theorem A.2 (Coherent-cluster extension)**.** Lean: `native_coherent_cluster_returns`, `native_coherent_cluster_expansion` [`ClusterExpansion.lean`]

The conclusions of Theorems 4.3 and 5.5 extend to every εP-coherent cluster, with returns and target lower endpoints on its complement. In the return statement one may take

$$N_c = \lfloor 160akC^2 \rfloor + 1, \quad \varepsilon = (128akN_c^2)^{-1}, \quad \eta = (8N_c)^{-1},$$

and sufficiently small $\gamma > 0$, with the controller budget $q \leq P/(64N_c^2)$. For cut expansion fix q and choose a sufficiently small positive cut-density constant. The exceptional count covers all clusters, signs and charts simultaneously, uniformly over $b \geq 1$.

Outline of the formal proof. The checked `native_cluster_slab_geometry` applies the bounded-speed slab selection to the actual append times and native head walks. The carrier is an integer walk of speed at most m. A displacement of size v crosses slab walls for at most v offsets, so offset averaging bounds the number of changes by $m(T - 1)/W < N_c$. The published short/long slab dichotomy depends on this change count, not unit speed of the carrier, and the raw-head stripe estimate is unchanged.

Preserve all protected cores between selected runs and refresh all fringes at their entries. In the short case all cores start blank. In the long case pay at most $a(W + m)$ symbols' bits at the outer start. The same calculation as Eq. (5) gives saving at least $P/(4N_c) - 3N_c db - a(k + 1)$. Each run additionally pays at most $ak \max(4j + 2, 2H + 1)$ bits for clustered fringes and noncluster windows, plus its controller. The native replay induction preserves every core together with the selected control

states; matching paid records then recover each demanded output bit. The typed noncluster imports inject into original completed complementary returns. Thus `native_bad_cluster_return_bound` derives the finite input count from the actual source records. Core ownership and current-version replay also prove the localized cut variant: a latest core writer after the outer start is in the selected region, and a fringe version is supplied by its current entry snapshot unless its last visit lies in the current run. Other cases are those in Section 5. The checked `native_bad_cluster_cut_bound` charges the complete local cut record and omits the import fields entirely; no small-return premise is needed for this cut variant.

There is no description of the real functions. A visited carrier slab has $|\ell| \le kT_* + W + 1$. At a selected time, chart endpoints differ from the corresponding oriented head position by at most $j + W$, so their integer roundings lie within $B_* = kT_* + W + 2j + 2$. A safe base geometry menu is

$$3^k(N_c+1)(2kT_*+W+2)(2B_*+1)^{4k}(T_*+1)^2[(T_*+1)^4(2T_*+1)^{k+3}]^{N_c}.$$

The 3^k choices encode exclusion or either orientation of each head. Consequently its logarithm is $O(\log P)$ at the chosen scale, despite input-dependent charts. The complete finite exponent is

$$P - P/(4N_c) + 3N_c db + a(k+1) + aN_c k \max(4j+2, 2H+1) + N_c q + \log_2 \mathcal{M}_c + \log_2 \mathcal{S}(kT_*, J_0, A).$$

Put $j = \lceil \varepsilon P \rceil$, $H = \lfloor \varepsilon P \rfloor$, $J_0 = \lfloor \gamma P \rfloor$. Windows/fringes and imports each cost at most $P/(32N_c)$, apart from constants, and the remaining terms cost at most $P/(32N_c)$ at large d under the stated state budget. For cuts replace the sparse-import term by the menu in Eq. (9) with N_c in place of N and choose its cost below $P/(32N_c)$. The checked `cluster_return_rates` and `cluster_cut_rates` choose one dimension threshold before all positive widths; the resulting finite exponent is strictly below $(1-\eta)P$. The demand upper bound uses injectivity of actual append times, and the lower bound uses the snapshot-adjusted geometric margin. The expansion endpoint covers every valid actual chart slab and all $H' \le H$ simultaneously, with cuts in the full original execution graph, including target lower endpoints outside the selected chunk. An all-head cluster has empty complement and is excluded on the same return good-input event. The checked singleton chart specializes the event to every free head; `native_cluster_two_return_heads` and `native_cluster_one_head_bad` recover the two-head and one-head consequences of Theorem 4.3 at these cluster constants. □

Lemma A.3 (Finite order-conflict chart)**.** Lean: `finite_order_chart`, `order_conflict_le_two_error` [`OrderChart.lean`]

For $m \ge 1$ and a nonempty finite family of vectors $v(t) \in \mathbb{R}^m$, put $z(t) = \sum_h v_h(t)$ and $D = \max_{z(t)\le z(u),h}(v_h(t) - v_h(u))_+$. There is a normalized chart with $-(m-1)D \le f_h(z(t)) - v_h(t) \le D$. Conversely a chart of error J implies $D \le 2J$.

Outline of the formal proof. Take coordinatewise prefix maxima over all vectors of sum at most $z(t)$. These vectors $g(t)$ form a coordinatewise chain, even when sums tie, and $v_h(t) \le g_h(t) \le v_h(t) + D$. Put $\Gamma_t = \sum_h g_h(t)$. The formal construction interpolates this chain by a continuous monotone max-envelope curve. Its hth coordinate is

$$G_h(x) = \max_t [g_h(t) - (\Gamma_t - x)_+] + \mathbf{1}_{h=h_0}(x - \max_t \Gamma_t)_+,$$

where h_0 is any fixed coordinate. It passes through $g(t)$ at $x = \Gamma_t$, and its coordinate sum is surjective onto $\mathbb{R}$. Reparametrize by that sum. Monotonicity makes the reparametrization well-defined even on constant intervals, and the normalized coordinate sum makes every coordinate 1-Lipschitz. Since $z(t) \le \Gamma_t$, the error at $z(t)$ is at most D in each coordinate. The errors sum to zero, giving the lower bound. Conversely monotonicity and two error allowances bound every ordered conflict by $2J$. □

For two work heads, Theorem A.2 and Lemma A.3 require macroscopic same-direction and macroscopic opposite-direction displacements, possibly on different time intervals: use orientations $(+,-)$ and $(+,+)$. They do not assert that every two-head path has a small-conflict cluster.

B Dynamic catalogues and prior-output assistance

A whole-word catalogue is a public finite set of possible tape words. It may depend on the machine, length and an entire prescribed input set, but not additionally on its individual input. The following stronger version allows indexing by the strict output prefix. Omitting that index gives the ordinary time-indexed catalogue theorem.

First enlarge the *analysis* model to a one-work-tape transition system whose rule receives the strict prior output string, with its length, in addition to its old native tuple. It receives no append timestamps, future bits, uncharged clock or mutable advice. This is not an output-reading native machine asserted to exist in the original model.

Lemma B.1 (Prefix-assisted one-head bound)**.** Lean: `Enlarged.prefix_finite_bound` [`PrefixFinite.lean`]

The uniform tail is `Enlarged.prefix_macroscopic_tail` *in* ***`PrefixTail.lean`****. The one-head case of Eq. (6), with no windows or imports, and its small-controller exponential success bound remain valid for this prefix-assisted analysis model.*

Outline of the formal proof. Keep the same slab, geometry, exterior of $U = \pi(O) \setminus I$, entry states and optional initial slab snapshot. Cache each replayed output by its ordinal. Immediately before a retained transition with output count o, every ordinal $j < o$ outside O is supplied by the exterior bit $X_{\pi(j)}$, because $\pi(j) \notin U$. Every $j < o$ inside O occurred at an earlier retained step and is in the cache. Chronological run order and paid entry/exit counters guarantee that no such ordinal belongs only to a future run. Thus the entire strict prefix is known *before* applying the rule. The free slab still persists, raw reads are exterior, and the common-old-tuple induction in Theorem 3.3 applies to the enlarged rule. No new message field is required. The geometric saving and state budget therefore give the identical finite and asymptotic bounds. The formal rule also permits public time and head coordinates, so it covers the stated interface as a special case without changing the message. □

Theorem B.2 (Dynamic prefix-catalogue handoff)**.** Not formalized; a complete proof is given below.

Fix a cap T, $n = 2T + 1$, and public catalogues $\mathcal{F}_{h,t,y} \subseteq \Gamma^{[-T,T]}$ of size at most $K \geq 1$, for every head, reached time and possible strict prefix of length at most P. Let G be any input set whose source runs halt by T with at most P appends and have at most one tape outside its current catalogue at every reached configuration, including halt. Put

$$r = \lceil \log_2(2k(T+1)K) \rceil, \qquad B_y = k(T+1)n^{k+1}(4T+1)(1+n^2)(1+T(P+1)KnA).$$

One prefix-assisted one-head system is exact on at least $|G|/2$ inputs, with at most $1 + QB_y 2^{kr}$ states and the same alphabet, transition count, raw moves and append times/bits. For fixed k, a, C, with $N_1 = \lfloor 160aC^2 \rfloor + 1$, $\rho = 1/(64N_1^2)$ and $\eta = 1/(8N_1)$, if

$$q \leq \rho P/2, \qquad \lceil \log_2 K \rceil \leq \rho P/[4(k+1)],$$

then at large d, uniformly over $b \geq 1$, fewer than $2^{1+(1-\eta)P}$ timely transpose-correct inputs satisfy the promise.

Proof (not formalized). Choose a binary linear hash with r output bits. For a fixed original input, compare each actual word at each reached (h, t) with each different member of its current catalogue, using the actual original strict prefix. There are at most $k(T+1)K$ comparisons, each colliding with probability 2^{-r}. The entire original run is independent of the hash. Averaging selects one hash collision-free on at least half of G. No union over hypothetical prefixes is involved. On these good inputs a matching hash identifies exactly the actual current catalogue word; aliases among unattained words are harmless. The common hash and surviving input set are checked by `exists_binary_common_half`; its comparison families are fixed before the hash is chosen. The native catalogue handoff remains written-only.

Maintain k live hashes with the local update rule from Theorem 6.1. Use the same active physical tape, translated coordinates and visited interval. Inactive words are decoded from their current catalogues. The active frozen word is initially blank, or is described by

$$(e, o_e, i, z, v), \quad 0 \le e < T,$$

meaning member i of $\mathcal{F}_{f,e,y[0:o_e)}$ with cell z replaced by v. Here o_e is the *pre-append* ordinal. The current strict prefix always contains that old prefix as a slice. There are at most $1 + T(P+1)KnA$ such frozen codes.

Initialize the active index to the unique catalogue excluding blank at time zero, or any index if none does; if two exclude blank the promise set is empty. Reconstruct the complete old tuple, apply the source rule, update all hashes and compute the newly appended bit. Only then test membership in the time-$t+1$ catalogues at the resulting prefix. If the active index changes to g, that tape was inactive before the step, so its postword is its known pre-catalogue member plus the single old-position write. Freeze this code with $(e, o_e) = (t, |y|)$ and reset the visited interval. This remains exact for a time-driven membership change, a preserving write or disjoint successive catalogues. There is one physical old-active write/move, as before.

Invalid table indices, ambiguous required decodes and out-of-range fields may halt. Good original runs establish the invariant, not a test for membership in G. Counting the displayed finite fields proves the state product. If $\lambda = \lceil \log_2 K \rceil$, its bit cost is $q + (k+1)\lambda + O_{k,a,C}(\log(P+1))$. The stated reserves fit ρP. Apply Lemma B.1 to the one-head analysis system and retain the factor two from hash selection. The original source still cannot read output; only the intermediate counting model has prefix assistance. □

Time-indexed catalogues give an ordinary native one-head simulator because its table need not consult a prefix. In either version the conclusion is per declared public catalogue system. It is not a union over all possible descriptions or an assertion that sparse Shannon entropy supplies a small support. Hamming balls about public templates are direct applications: on $2T+1$ cells their size is at most $\sum_{j \le m} \binom{2T+1}{j}(A-1)^j$. Sufficiently small fixed linear m fits the rate budget, as does $m = o(P)$.

C Locally reconstructible incoming values

The incoming record can pay only for failed predictions. The prediction interface must be fixed before selecting a slab, and must reconstruct the actual old value rather than a final version of the cell. Fix $p \ge 1$ integer raw ports $v_j(t)$, each moving by at most one per successful source transition and starting at a public position. One is the actual input head. A virtual port has a public update rule using the old source state, positions, complete old native tuple, the declared raw windows and the resulting native transition. All its auxiliary storage belongs to an augmented state set of size at

most 2^q. It has no other raw query or hidden history. Public time and the current append ordinal are available.

Fix one carrier f for the entire execution, a radius R, a finite label alphabet Λ of size at most 2^z, and public getters $G_{f,h}$ for $h \neq f$. A getter receives one whole-run label, the known old augmented state and positions, time and output ordinal, the carrier's old symbol, all radius-R raw windows, and the entire *strictly prior* output string. It receives no unknown sibling old symbol. These are offline analysis arguments, not additional native source instructions. In particular, the source still cannot read output, and the virtual-port update contract does not acquire output-prefix access. Getters may be nonlinear.

For a chosen carrier and label, count an error at every actual noncarrier consecutive-return endpoint whose excursion exceeds H and whose old symbol differs from the getter on its actual old context. This statistic is fixed on the whole run before choosing its replay region.

Theorem C.1 (Replay with local getters)**.** Lean: `Getter.native_fixed_geometry_bits_bound` [`GetterGeometry.lean`]

Fix a public output permutation π and a selected slab as in Section 3, with r retained runs. Record, additionally, each port's entry position and run extrema, and each noncarrier head's global successful visit interval strictly before each run. Let I be the union of the port extremum intervals enlarged by R, restricted to $[0, P)$; let O be the retained append ordinals and $U = \pi(O) \setminus I$. For fixed geometry and at most J selected getter errors the number of represented correct inputs is at most

$$2^{P-|U|+z+rq} A^{\iota W+(k-1)r(2H+1)} \mathcal{S}((k-1)T_*, J, A), \qquad \iota = \mathbf{1}_{a_0>0}.$$

If there are at most J whole-run errors on off-carrier returns of excursion greater than H, the selected errors number at most J.

Outline of the formal proof. The error bound is `Getter.error_support_le_returns`. A successful unit walk visits every integer between its extrema. Thus its global pre-entry visit set is empty or one integer interval; record an empty flag or its endpoints, excluding the arrival position unless previously scanned. The enlarged port extrema cover every raw value used in a retained transition, including the native read and the virtual-port update.

Use the message of Theorem 3.3, with one label and complete augmented entry states. At a first in-run read outside the entry window, first test the recorded past interval. Outside it the cell is virgin and blank. Inside it use a tag carrying the actual old symbol exactly when the getter is wrong, and the getter otherwise. A tag is a genuine long return: its last global visit precedes the run, whereas the run's entry time lies strictly between that visit and the read and is farther than H from the cell. The correspondence preserves the typed pair (t, h) and is injective. Virgin cells need no tag even when their getters are wrong.

Keep the carrier slab dictionary across all retained runs, as before, and keep a second dictionary of already replayed output bits by ordinal. Just before a retained step with ordinal o, every earlier ordinal $j < o$ outside O has $\pi(j) \notin U$, so its bit is supplied by the exterior input. Every $j \in O$ with $j < o$ was appended at a strictly earlier retained step and is in the second dictionary. This supplies the strict prefix before any unknown sibling symbol is reconstructed. Outputs in a skipped gap have ordinals outside O; their timestamps need not be known.

At each run install the paid state, positions, counter and entry windows. For each noncarrier read use, in order, its most recent in-run write, its entry-window value, virgin blank, or its tag/getter. In the last case the getter's arguments are already determined from the common old context, carrier dictionary, exterior raw bits and prior-output dictionary. A missing tag therefore certifies the correct prediction on this actual input. Only after reconstructing the whole old tuple apply simultaneous

old-position writes, moves, the optional append and the declared port updates. The usual induction reproduces every retained transition. Unretained carrier writes cannot touch the slab; skipped noncarrier history is paid at the next entry.

Correctness recovers X on $\pi(O)$ and hence U. Together with the exterior, comparing two actual runs with the same message gives equal inputs, even across different fine traces. No decoder on invalid records is needed. Count the same joint typed tag/value menu as in Lemma 3.2; the label is paid once. No independence is used. □

Theorem C.2 (Macroscopic local discrepancies)**.** Lean: `Getter.native_finite_getter_bound` [`GetterFinite.lean`]

The uniform tail is `Getter.native_macroscopic_getter_tail` *in* ***`GetterTail.lean`****; existence of the stated density is* `Getter.exists_getter_density` *in* ***`GetterDensityExistence.lean`****. Fix $k, p, a, C \geq 1$, $A \leq 2^a$, and one declared getter system. Put $T = \lfloor C(P+1) \rfloor$ and*

$$N = \lfloor 640apC^2 \rfloor + 1, \quad W = \lceil P/(320apC) \rceil, \quad R \leq \lfloor P/(8pN) \rfloor.$$

For $P \geq 4p$, the inputs transposed by T for which some fixed carrier and some label have at most L errors on returns of excursion greater than H number at most

$$\begin{aligned} 2^{P-P/(4N)+E}, \qquad E &= 3Ndb + a + a(k-1)N(2H+1) + Nq + z \\ &\quad + \log_2 \mathcal{M}_p + \log_2 \mathcal{S}(kT, L, A), \\ \mathcal{M}_p &= k(N+1)(2T+W+2)(T+1)^2 \\ &\quad \cdot [(T+1)^4(2T+1)^{k+3p}(1+(2T+1)^2)^{k-1}]^N. \end{aligned}$$

Set $\varepsilon = [128a \max(1, k-1)N^2]^{-1}$, $\eta = (8N)^{-1}$, and choose $0 < \gamma \leq 1/2$ with $\gamma[a + \log_2(2ekC/\gamma)] \leq 1/(64N)$. If $q \leq P/(64N^2)$ and $z \leq P/(64N)$, then, uniformly over every $b \geq 1$ at sufficiently large d, fewer than $2^{(1-\eta)P}$ timely correct inputs have at most $\lfloor \gamma P \rfloor$ such errors at $H = \lfloor \varepsilon P \rfloor$.

Outline of the formal proof. Offset averaging gives a width-W grid with fewer than N slab changes, since $T/W \leq 640apC^2 < N$. Apply the chronological slab dichotomy with cap $B = \lfloor P/(4p) \rfloor$. For completeness, if at least half the outputs lie in slabs with at most B visits, one of at most N slabs has at least $P/(2N)$ outputs and starts blank. Otherwise split each long slab's $n_s > B$ visits into $q_s = \lceil n_s/B \rceil$ consecutive groups. Each is the slab intersected with a time interval, and $q_s B \leq n_s + B - 1 < 2n_s$. Thus $B \sum_s q_s \leq 2T$; some chunk has $PB < 4T|O|$. Both branches have at most B selected times and N runs.

The sum of lengths of all enlarged port intervals is at most $p|C| + 2pRr \leq P/2$. Their pr intervals meet at most $P/(2db) + 2pr$ raw rows. At least $F \geq d/2 - 2pr$ rows are untouched. Each of the r output intervals demands at least its length divided by d, less $2b$, from each such row. Consequently

$$|U| \geq F|O|/d - 2rbF \geq |O|/2 - 2pr|O|/d - 2rdb \geq |O|/2 - 3rdb,$$

using $F \leq d$ and $|O| \leq B \leq P/(4p)$; the inequality also holds when a displayed lower bound is negative. The short branch saves at least $P/(4N) - 3Ndb$ bits and pays no slab. In the long branch, $B \geq P/(8p)$ and $T \leq 2CP$ give $|O| > P/(64pC)$, while $aW \leq P/(320pC) + a$. Hence in both branches

$$|U| - a\iota W \geq P/(4N) - 3Ndb - a.$$

The geometry menu records the carrier, slab endpoint, two outer times and run count. Each run pays four times/counters, k work entries, p port entries, $2p$ port extrema and the $k-1$ past intervals. Port positions are encoded relative to their public origins, with $2T+1$ choices each; absolute origins

need not be bounded. This gives $\mathcal{M}_p$ exactly as a safe overcount. Choose one admissible witness by a fixed finite rule, apply Theorem C.1, and sum over the menu to obtain the finite bound.

For $L \le \gamma P$, put $x = \gamma/(2kC2^a) \in (0,1]$. The binomial identity gives $x^L \mathcal{S}(kT, L, A) \le (1+Ax)^{kT}$. Taking logarithms, using $\log(1+Ax) \le Ax$ and $kT \le 2kCP$, bounds the bit cost by $\gamma[a + \log_2(2ekC/\gamma)]P \le P/(64N)$. This argument includes $L = 0$ without dividing by L. The H-dependent part, state and label each cost at most $P/(64N)$. Finally $3Ndb + a + a(k-1)N + \log_2 \mathcal{M}_p = o(P)$ uniformly in $b \ge 1$, since $db = P/d$ and $P \ge d^2$; take it at most $P/(64N)$. The exponent is at most $(1 - 11/(64N))P < (1-\eta)P$. The threshold is uniform in transition tables and public getters with the declared resources, but the event is per fixed system, not their union. □

C.1 One shared set of spatial checkpoints

Let a public finite family of deterministic scans have finite state spaces $S_{\nu,i}$, public initial states, and updates $s_{\nu,i+1} = F_{\nu,i,X_i}(s_{\nu,i})$. No invertibility is required. For a public partition $0 = a_0 < \cdots < a_K = P$ with gaps at most $R+1$, record the states at internal anchors. Their joint support is at most

$$2^Z, \qquad Z = \sum_{\nu} \sum_{0<j<K} \log_2 |S_{\nu,a_j}|.$$

Its actual joint image can replace this product. Input-dependent initial states, terminal states exposed outside the valid domain, and selectors must also be included in the complete label.

Lemma C.3 (Shared scan checkpoints)**.** Lean: `scanState_window`, `preceding_residue_anchor`, `checkpoint_residue_average`, `checkpointLabel_card`, `checkpoint_image_card_le` [`ScanCheckpoints.lean`]

Equal anchor states and equal raw radius-R windows at i, where the preceding anchor is at most R positions earlier, give equal states at i and $i+1$, including position-dependent state types. The joint label alphabet has cardinality $\prod_\nu \prod_{0<j<K} |S_{\nu,a_j}|$, and its attained image is no larger. For any public real weights w_i there is a residue partition with gaps at most $R+1$ whose internal anchors satisfy

$$(R+1) \sum_{i \text{ internal anchor}} w_i \le \sum_{i=1}^{P-1} w_i.$$

Outline of the formal proof. For the preceding anchor $a_j \le i < a_{j+1}$, start at its recorded state (the public initial state when $j = 0$) and apply the original scan updates at $a_j, \dots, i-1$. These coordinates lie in $[i-R, i)$, and induction recovers the actual state. The update using X_i gives the next state. Thus the same label works under arbitrary later port reversals, without an inverse or a source scan observer. Position-dependent state types cause no difficulty: each update has its declared source and target type.

Partition at $0, P$ and at internal positions in one residue class modulo $R+1$. All gaps are at most $R+1$. Every internal position appears in exactly one of these $R+1$ menus, so a minimum-weight residue has at most their average weight. Its choice uses only public state cardinalities. The formal preceding-anchor construction is zero when i precedes the chosen residue r, and otherwise $r + (R+1)\lfloor (i-r)/(R+1) \rfloor$. The dependent-function cardinality formula gives the label product; image cardinality is bounded by its ambient finite type. □

For $w_i = \sum_\nu \log_2 |S_{\nu,i}|$, the inequality gives $Z \le \sum_i w_i/(R+1)$. At $R = \lfloor P/(8pN) \rfloor$, equal-length blocks have $\lfloor (P-1)/(R+1) \rfloor < 8pN$ internal anchors. Hence a common state-bit bound $\sum_\nu \log_2 |S_\nu| \le P/(512pN^2)$ fits $Z \le P/(64N)$. The alternative profile bound $\sum_i w_i \le P^2/(512pN^2)$ fits by $R+1 > P/(8pN)$ and averaging.

Substitution in Theorem C.2 gives the same discrepancy tail with $z = Z$. The description must hold throughout the actual run, including preparation, or preparation errors must be counted. An offline scan getter does not itself supply a fast native producer. A reversible scan has the simpler radius-one implementation: retain its prefix state at each port, apply the update on a forward move and its inverse on a backward move, charging those observer states to q. This includes noncommutative finite group prefix products.

Proposition C.4 (Paid finite-state archive preparation)**.** Lean: `ScanLoader.native_load`, `ScanLoader.native_resources`, `ScanLoader.native_ready_properties` [`ScanLoader.lean`]

The complete success bound is `ScanLoader.native_scan_loader_tail` *and* `ScanLoader.native_scan_loader_exponential` *in* `ScanLoaderTail.lean`*. Fix r finite-state scan codes with updates $F_j : S_j \times \{0,1\} \to S_j$ and symbol rules $D_j : S_j \times \{0,1\} \to A_j$. A uniform native producer loads their aligned code words on r blank archive tapes, resets all heads to zero, leaves a further carrier tape blank and output empty, in exactly $2P + 4$ steps. With a consumer having Q_C states, the composed controller has at most $Q_C + \prod_j |S_j| + 3$ states over one fixed enlarged alphabet. If every consumer instruction repeats each archive's old scanned symbol in that archive's write, but otherwise has arbitrary adaptive head motion and an unrestricted carrier, its complete linear-time transposition success set has the exponential bound of Theorem C.2, uniformly over all positive widths.*

Outline of the formal proof. Choose distinct blank, left marker, right marker and data encodings. In one START step move archive heads from 0 to -1, preserving blank. In MARK write left markers at the old positions and return to 0. Keep raw and carrier heads at 0. In LOAD, the ith raw bit and all archive heads are at i; their scanned cells are virgin. Write $D_j(s_j(i), X_i)$, update all s_j from the same old bit and move raw/archive heads right. At raw EOF at P, write right markers and move left, discarding the scan state. REWIND preserves the P data cells and moves left to -1; at that raw endmarker it preserves the markers, moves right and enters the consumer. The count is $2 + P + 1 + P + 1$. No length counter is used. The disjoint control sets are the consumer states, the product of scan states, and START, MARK, REWIND. All writes occur at old positions. The formal partial-archive invariant proves the exact contents and coordinates after each loaded bit; the rewind induction preserves the completed words. It includes empty inputs and zero archives. Fixed finite enumerations compile this table to the frozen native machine with exactly one native step per transition. The arbitrary consumer continues with no linking step, including its outputs and halting behavior, by `ScanLoader.native_prepare_then_consume` and `ScanLoader.native_consumer_halt` in `ScanLoaderConsumer.lean`.

Use the raw head and archive-head coordinates as $p = r + 1$ ports. During START, MARK and LOAD the getter returns blank at the actual scanned cells, including the EOF step. During REWIND and consumption it evaluates the completed marked code word using the shared checkpoints. Define a total public decoder by choosing any length-P word compatible with its label and local window, or returning blank when none exists. Scan locality proves that all compatible words give the same archive symbol, and the actual input is compatible. Thus this selection defines the getter; its correctness is a conclusion, not an additional hypothesis.

The reached-configuration invariant derives every archive old read, including markers and blank exterior. Each consumer archive write repeats the old symbol, so induction preserves the completed words. A one-state observer follows the actual archive-head moves. Consequently `ScanLoader.nativeArchiveGetters_zero_errors` in `ScanLoaderGetters.lean` proves that the original whole-run discrepancy set is empty, including preparation.

Put $K = 8pN$ and $R = \lfloor P/K \rfloor$. Since $P \le (R+1)K$, at most K checkpoint slots suffice; padding is included in the finite label. Their product alphabet has size $(\prod_j |S_j|)^K$, independent of P. Both its bit bill and the fixed controller's bill therefore meet the stated budgets beyond one dimension

threshold, uniformly over all positive widths. Apply Theorem C.2, with preparation included in the total linear cap. Its formal bound also includes $k = 1$, which covers $r = 0$ directly. □

Small scan state is compatible with a full-support nonlinear archive: $Y_i = X_i \operatorname{xor} s_i$, $s_{i+1} = s_i \vee X_i$, $s_0 = 0$, is bijective by successive recovery of X_i from Y_i, s_i, even though the state update at input 1 is noninvertible. This example supplies no transpose decoder. Nor does a final scan-code identity justify intermediate versions: overwrite an archive temporarily, read it, and restore it by another paid scan. Those intermediate reads need their own getter or tags.

D Changing carriers relative to input-dependent references

Public catalogues need not contain all raw-dependent words, even when each word has a simple local description. We therefore use a counted analysis model with one physical work tape, p continuous raw-window ports, the strict prior output, and one immutable input-dependent label. Its whole transition may use these arguments. Its finite control has 2^{q_*} states; time, positions and the append ordinal are explicit arguments, and every other evolving register is charged. It starts in a public state with blank work tape and public port origins. Every work write is at its old position, all heads and ports move by at most one, and at most one bit is appended. This model is only an intermediate counting device.

Lemma D.1 (Counting the entire enlarged transition)**.** `Lean`: `Enlarged.finite_bound` [`EnlargedFinite.lean`]

The uniform tail is `Enlarged.macroscopic_tail` *in* ***`EnlargedTail.lean`****. Let* $P = d^2b \geq 4p$, $T = \lfloor C(P+1) \rfloor$, $N = \lfloor 640apC^2 \rfloor + 1$, $W = \lceil P/(320apC) \rceil$, *and* $R \leq \lfloor P/(8pN) \rfloor$. *A fixed system of the preceding kind with alphabet* $\mathbb{F}_2^a$ *and label support at most* 2^z *correctly transposes, for some label, at most*

$$2^{P - P/(4N) + 3Ndb + a + Nq_* + z + \log_2 \mathcal{M}_1}$$

inputs, where

$$\mathcal{M}_1 = (N+1)(2T+W+2)(T+1)^2[(T+1)^4(2T+1)^{1+3p}]^N.$$

If $q_* \leq P/(64N^2)$ *and* $z \leq P/(64N)$, *this is less than* $2^{(1-1/(8N))P}$ *uniformly in* $b \geq 1$ *at sufficiently large* d.

Outline of the formal proof. Apply the slab selection and raw-footprint calculation in Theorem C.2 to the physical work head. The selected chunk has at most N runs and saves $P/(4N) - 3Ndb - a$ bits after its complete initial slab is paid. There are no noncarrier symbols or error tags. Record run-entry control, port positions/extrema, physical work position, times and append counters, and one label. This is $\mathcal{M}_1$. Replay the entire transition from the persistent work slab and the known raw windows. Assemble its strict prior output exactly as in Theorem C.1; this occurs before applying the transition, so no current output bit is assumed. The same induction applies to the whole enlarged rule, including its port updates. Correctness recovers the omitted input. The label is paid once and run-entry controls at most N times, giving the stated finite bound. The two linear budgets and $3Ndb + a + \log_2 \mathcal{M}_1 = o(P)$ leave the claimed slack uniformly in b. □

Fix $k \geq 2$ native work tapes, $\ell \geq 1$ references per tape, and an embedding of the source alphabet in $V = \mathbb{F}_2^a$ with blank zero. Put $E = [-T, T] \cap \mathbb{Z}$, $n = 2T + 1$. Each reference uses a public 1-Lipschitz map $\phi_{hi} : \mathbb{Z} \to \mathbb{Z}$ and a public local function γ_{hi} of the coordinate and the radius-r_0 raw window centered at $\phi_{hi}(z)$. One optional immutable seed $\lambda_0 = f(X)$ is fixed before the hash used below. Its joint support on the fixed promise set is at most 2^{Z_0}. An observer with U states contains

all cutoffs $b_{hi} \in [0, P]$ and auxiliary storage, starts publicly, and changes each cutoff by at most one. Its updates may use the seed, old source/observer context, strict output, actual old source symbols and the declared old raw windows, but no other individual raw data. Initial cutoffs and port origins are public, independent of the seed. All rules are independent of the later hash choice.

Let $D_{hi}(t, Y, \lambda_0) \in V^E$ be any public dense mask. At reached time t define

$$F_{hi}(t, X)(z) = \mathbf{1}_{[0,b_{hi}(t))}(z)\gamma_{hi}(z, \mathrm{window}_X(\phi_{hi}(z)), \lambda_0) + D_{hi}(t, Y_t, \lambda_0)(z),$$

where Y_t is strictly prior output. Words are zero outside E. Write

$$K_s = \sum_{j=0}^{\min(s,n)} \binom{n}{j}(2^a - 1)^j, \quad r = \lceil \log_2(2k(T+1)\ell K_s) \rceil, \quad p = 1 + 2k\ell.$$

Theorem D.2 (Raw-relative carrier handoff)**.** Not formalized; a complete proof is given below.

Let G be any promise set of source inputs whose runs halt by T, append at most P bits, obey the observer promises, and at every reached configuration, including halt, have at most one work word farther than s cells from every current reference on its tape. One common analysis system of Lemma D.1, with p ports and radius $r_0 + 1$, reproduces the source exactly on at least $|G|/2$ inputs, with the same transition count, raw input trajectory and append times/bits. Its state and label bills are

$$Q_* \le 2 + QUB_{\mathrm{ref}}2^{k(\ell+1)r}, \qquad z \le Z_0 + k\ell r,$$

where Q is the number of source states and

$$B_{\mathrm{ref}} = k(T+1)n^{k+2}(4T+1)(1+n^2)(1+T(P+1)^2\ell K_s n 2^a).$$

No ordinary native one-tape simulation is asserted.

Proof (not formalized). Choose a binary linear hash $H : V^E \to \mathbb{F}_2^r$. For fixed $X \in G$, its original augmented run and seed are determined independently of H. Compare each actual word $w_h(t)$ with every different candidate $F_{hi}(t, X) + e$, where e has at most s nonzero cells. Each nonzero difference hashes to zero with probability 2^{-r}: in each independently chosen binary row of H, one nonzero coordinate determines a fair bit. There are at most $k(T+1)\ell K_s$ comparisons, so a union bound followed by averaging supplies one H good on at least half of G. This event refers only to original trajectories. Empty G is immediate. The same finite selection theorem `exists_binary_common_half` applies with $B = k(T+1)\ell K_s$. It proves the single-hash half-set claim, including empty comparison families; the native reference-system construction below remains unformalized.

Maintain k actual-word hashes $Z_h = H(w_h)$ and $k\ell$ raw-prefix hashes $L_{hi} = H(\mathbf{1}_{[0,b_{hi})}\gamma_{hi})$. Supply their initial raw values by one label together with λ_0; this costs at most $Z_0 + k\ell r$, and their evolving copies are separately in the controller. The actual hashes initially vanish. An old-symbol change $u \to v$ at y updates Z_h by $H(\mathbf{1}_{\{y\}}(u + v))$, even for dense words. A cutoff increment adds the hash contribution at its old boundary b; a decrement removes the contribution at $b - 1$. Contributions outside E are zero. Use continuous ports for the source input, every $\phi_{hi}(y_h)$ and every $\phi_{hi}(b_{hi})$. Their unit motion follows from the 1-Lipschitz maps. The old radius-$(r_0 + 1)$ boundary window contains the decrement's required radius-r_0 window. No port jumps at handoff.

Test, in a public fixed order, the finite equations

$$H(e) = Z_h + L_{hi} + H(D_{hi}(t, Y_t, \lambda_0)).$$

On a good original run every matching pair describes the actual word and an unmatched word has none. Several pairs describing the same word are allowed; choose the first. Candidate words outside the source alphabet may be included in this test, so no global raw-word legality query is needed. Only the reconstructed scanned symbols need belong to the source alphabet.

Keep the active index f, logical source positions, physical position p_0, offset u with $y_f = p_0 + u$, and an empty or integer interval I of physical positions scanned in previous transitions of this carrier phase. Its frozen entry word is initially blank, or is encoded by

$$(e, o_e, b, i, e_s, z, v), \quad e < T, \quad o_e, b \in [0, P], \quad e_s \in \mathrm{Err}_s,$$

meaning the old reference with cutoff b, mask $D_{fi}(e, Y_t[0 : o_e), \lambda_0)$ and error e_s, with cell z *overwritten* by v. For $j \in I$ the physical symbol is the current logical symbol at $j + u$; outside I the logical word still equals this frozen entry word. Its fresh symbol uses the logical-head raw port and the historical prefix slice, never the current cutoff or mask.

In a public INIT state install the public source/observer data, initialize hash registers from the label and choose the unique unmatched tape, or any tape if all match. This initialization is part of evaluating the first ordinary transition, with no extra step or free tape write. Initially all logical words are blank, $u = 0$ and I is empty. Invalid data can halt.

At a represented step decode every inactive scanned symbol from a matching pair. Decode the active symbol physically when $p_0 \in I$, and from its frozen code otherwise. Having the entire old tuple, apply the source rule, observer update and the local hash updates. A source halt causes no write or append. Otherwise the sole physical action writes the old active tape's new symbol at p_0 and follows that tape's chosen move. All logical positions and raw ports make their unit moves, and the same bit is appended at the same step. Only now test the post-step references using the new prefix.

If the active tape stays selected, extend I by the old physical position; unit motion keeps it an interval. If a new carrier g is selected, it was inactive before this transition. Freeze its PRE matching pair, old cutoff, time and output ordinal, together with its one actual old-position write. This describes its exact POST word. Set the new offset from its new logical position and the already moved physical head, and reset only I to empty. The former carrier is now matched. Stale physical cells are ignored on first visit and overwritten with their correct next values; no clearing or seek is performed. These statements establish the invariant by induction through arbitrary switches, including changes caused solely by masks or cutoffs.

Malformed fields, nonunit moves or unavailable matches can halt; no branch tests membership in the good set. The invariant proves exactness on that set with its true label. Count source/observer state, active index, time, $k + 2$ positions in $[-T, T]$, offset in $[-2T, 2T]$, interval, the $k(\ell + 1)$ hashes and the frozen code. Their product is the displayed bound, with INIT and a rejecting state added. Port positions are derived from counted logical positions/observer state; the strict prefix itself is an analysis argument. The immutable seed support can be counted on the fixed G before choosing H, with no second largest-fibre loss. □

Corollary D.3 (Two words far from all current references)**.** Lean (partial): `reference_tail_of_half_simulation`, `reference_cache_sparse_reserves` [`ReferenceTail.lean`]
Not formalized: Lean derives the complete state and label budgets, output correctness and success tail from one common native-prefix simulation on at least half the promise set, with the literal cardinality bills. The reference-carrier construction supplying that simulation remains unformalized.

Fix the above resources, set $N = \lfloor 640apC^2 \rfloor + 1$ *and* $D_0 = k(\ell + 1) + 1$*. If*

$$q_{\mathrm{src}} + q_{\mathrm{obs}} \le \frac{P}{128N^2}, \quad \lceil \log_2 K_s \rceil \le \frac{P}{256D_0N^2}, \quad Z_0 \le \frac{P}{128N}, \quad r_0 + 1 \le \left\lfloor \frac{P}{8pN} \right\rfloor,$$

then fewer than $2^{1+(1-1/(8N))P}$ *timely correct inputs satisfy the one-exception promise, uniformly for all* $b \geq 1$ *at large* d*. For a sufficiently small fixed* $\delta > 0$*,* $s = \lfloor \delta P \rfloor$ *meets the sparse budget. Outside that exception, a reached configuration has two distinct words each farther than* s *cells from every current reference for its tape.*

Proof (not formalized). Put $\lambda_s = \lceil \log_2 K_s \rceil$ and $L_0 = \lceil \log_2(2k(T+1)\ell) \rceil$. Then $r \leq \lambda_s + L_0$, and $B_{\mathrm{ref}} \leq K_s B_0$ for the same polynomial factor with K_s removed. The compiler therefore has

$$q_* \leq q_{\mathrm{src}} + q_{\mathrm{obs}} + D_0 \lambda_s + O(\log(P+1)) \leq P/(64N^2)$$

eventually, uniformly in b. The same estimates give $k\ell r \leq P/(128N^2)$, hence $z \leq P/(64N)$ including Z_0. Apply Lemma D.1 and retain the factor two from the common-hash half-set. For the last assertion use $n \leq \lceil 4C+1 \rceil P$ and $\log_2 K_s \leq s \log_2(en(2^a-1)/s)$ for sufficiently small s/n. Its ratio to P tends to zero with s/P; choose a fixed positive δ with enough slack for the ceiling. Negating the one-exception promise gives the stated simultaneous witness. This is per declared system, not a union over input-selected reference rules. The checked budget includes both administrative states and every integer ceiling. It uses a single dimension threshold before the width, source, observer and enlarged transition table are chosen. □

D.1 Static scans as input-dependent references

Take $L = \lfloor P/(8pN) \rfloor$. The checkpoint construction of Lemma C.3 with gaps at most L evaluates each static original-raw scan using radius $L-1$; cutoff decrements then use total radius L. Its one joint seed is reused at every handoff. It fits the preceding corollary when its support charge is at most $P/(128N)$. In particular, for $P \geq (8pN)^2$ the number of equal-grid internal anchors is $\lfloor (P-1)/L \rfloor \leq 8pN$: writing $P = 8pNL + r$, $0 \leq r < 8pN$ and $L \geq 8pN$ proves the inequality. Thus common state bits $w \leq P/(1024pN^2)$ suffice. Public offset averaging also gives the sufficient profile condition $\sum_i w_i \leq LP/(128N)$.

The scan descriptions remain functions of the original raw input; the current cutoffs, masks and sparse deviations may evolve. A large offline scan table has no native production guarantee. Public catalogues with a growing number of references retain their separate scope, as do fixed-carrier local discrepancies. Neither conclusion supplies independence, persistence or output usefulness of the deviations. A raw-copy track alongside an arbitrary payload does not make that payload a small residual.

E Secondary modification episodes

Keep the reference and observer interface of Appendix D. A tape is *matched* when its current whole word differs in at most s cells from some current reference. Define a canonical carrier on the original run: initially choose the first unmatched tape, or tape zero if all match; subsequently retain the carrier if it is unmatched, otherwise choose the first unmatched tape, retaining the old carrier if all match. This rule is independent of the hash used in the analysis. Whenever any tape is unmatched, the carrier is unmatched.

On a run with τ successful transitions, an inactive unmatched episode of tape h is a maximal interval $I = [u, v)$ of reached configuration indices $0, \ldots, \tau$ on which h is unmatched and is not the carrier. Define

$$D_h(I) = \{ y_h(t) : t \in I,\ t < \tau,\ \text{the old-position write at } t \text{ changes the symbol} \}.$$

Addresses, not write occurrences, are counted. A restore still counts its address. A successful exit transition is included, the entering transition $u-1$ is excluded, and the halting configuration contributes no write.

Lemma E.1 (Changed cells and a frozen word)**.** Lean: `ExecutionPrefix.unchanged_outside_changedCells`, `ExecutionPrefix.residual_card_le`, `ExecutionPrefix.residual_card_le_entering_add_episode`, `ExecutionPrefix.changedCells_card_le_duration` [`HistoricalCache.lean`]

In an actual successful execution prefix, let $D(u,v)$ be the distinct old addresses changed on one tape in steps $[u,v)$. Outside $D(u,v)$ its value at v equals its value at u. In any finite address window, the number of cells differing from a fixed word at v is at most their number at u plus $|D(u,v)|$. In particular, an initial residual of at most s cells followed by one entering step and at most m distinct changed addresses in $[u+1,v)$ leaves at most $s+m+1$ residual cells. Also $|D(u,v)| \le v-u$.

Outline of the formal proof. Induct through actual successful transitions. A cell different from the old head position is preserved; at that position it is preserved unless the step belongs to the changed-cell set. The final residual is contained in the union of its initial residual and this set. Split off the first old position and use finite union cardinality. Finally the changed-cell set is the image of a subset of $[u,v)$, giving the duration bound. The set records actual symbol changes, so identity writes are excluded and repeated changes or restores use only one address. □

Theorem E.2 (Historical reference caches)**.** Not formalized; a complete proof is given below.

Let G satisfy the source/observer, halt and output-length promises of Theorem D.2, replacing its one-exception promise by $|D_h(I)| \le m$ for every episode. Put $M = s+m+1$, retain K_j, r, p from that theorem, and set

$$B_{\mathrm{cache}} = k(T+1)n^{k+2}(4T+1)(1+n^2)[1+(T+1)(P+1)^2\ell]^k.$$

An analysis system exact on at least half of G has the same time, input trajectory and append times/bits, with

$$Q_* \le 2 + QUB_{\mathrm{cache}}K_M^k 2^{k(\ell+1)r}, \qquad z \le Z_0 + k\ell r.$$

It uses one blank physical work tape and the same continuous ports and radius as before. There is no restriction on episode length or count, or on the number of simultaneous unmatched tapes.

Proof (not formalized). Choose the same common hash event as in Theorem D.2: compare only current actual words with current radius-s candidates on original trajectories. It gives the true matching decisions on at least half the inputs. No historical radius-M search is included in the event.

Give every tape a cache consisting of a frozen base, either BLANK or (e,o,b,i), and an explicit error $\epsilon \in \mathrm{Err}_M$. At current prefix Y, the latter base denotes

$$\mathbf{1}_{[0,b)}\gamma_{hi}(\,\cdot\,, \mathrm{window}_X, \lambda_0) + D_{hi}(e, Y[0:o), \lambda_0).$$

The inactive caches describe current words; the active cache describes its frozen phase-entry word, with subsequent active writes represented physically. An inactive matched tape is refreshed from its first current matching pair at every positive time. Initially every cache is BLANK with zero error. An inactive write $u \to v$ at y changes its error by $\mathbf{1}_{\{y\}}(u+v)$, leaving its base fixed.

We verify the support bound at the episode boundary. If an episode starts at $u > 0$, its tape was not active at $u-1$: an unmatched old carrier at u would have been retained. Maximality therefore says it was inactive and matched at $u-1$. Its radius-s refreshed error (or exact blank error if

$u-1=0$) acquires at most the one entering address. Subsequently its support is contained in those old error addresses, that entering address, and $D_h(I)$. Thus it is at most $s+1+m$, through every successful exit write as well. An episode starting at zero needs only $D_h(I)$. A matched inactive tape promoted immediately after one write needs at most $s+1$. These cases cover every inactive cache before refresh or promotion.

Retain the lazy physical invariant, logical positions, current hashes and continuous ports from Theorem D.2. Reconstruct inactive symbols directly from their caches, even when currently unmatched. The active symbol comes from the physical tape on its visited interval and from its frozen cache elsewhere. All queries use the current logical-head raw ports and the same old windows/seed. Reconstruct the complete old tuple, then compute the source rule, observer/hash updates and tentative error updates for every old inactive tape. Keep the old active cache frozen.

After computing the optional append, test current matches and choose the new carrier by the canonical policy. If it stays, extend the visited interval. If it changes to g, freeze g's exact tentative POST-write cache, set the new offset, and reset the visited interval. The previous carrier must now be matched, so refresh its inactive cache from its new matching pair; refresh all other matched inactive tapes, and preserve the remaining updated historical caches. Only the old carrier makes the one physical write and move. The support argument proves that every cache needed at a switch is legal, including a blank base with nonzero error.

INIT installs public original data, zero actual hashes and blank caches, and obtains initial raw hashes from the one joint seed/hash label, within the first transition evaluation. Multiple unmatched tapes at time zero cause no problem because all actual words are blank. On invalid branches finite range/support checks can halt. The invariant proves actual source semantics through halt on the good inputs without testing the promise. No episode counter, old-reference search, fresh label, clearing or seek is needed. Each of the k caches has $[1+(T+1)(P+1)^2\ell]K_M$ descriptions. Multiply by the same state, observer, head, interval and hash fields as before to obtain the claimed product, including the active frozen slot. The label bill is unchanged. □

Corollary E.3 (Many distinct secondary modifications)**.** Lean (partial): `cache_tail_of_half_simulation`, `reference_cache_sparse_reserves` [`ReferenceTail.lean`]
Not formalized: The exact budgets and success tail are checked from one common native-prefix half-set simulation with the stated cardinality bills. The historical-cache compiler and its canonical-episode invariant remain unformalized.

Put $N=\lfloor 640apC^2\rfloor+1$ *and* $A_0=k(\ell+2)$. *Assume*

$$q_{\mathrm{src}}+q_{\mathrm{obs}}\le P/(128N^2),\quad \lceil\log_2 K_M\rceil\le P/(256A_0N^2),\quad Z_0\le P/(128N),\quad r_0+1\le\lfloor P/(8pN)\rfloor.$$

Uniformly for every $b\ge 1$ *at sufficiently large* d, *fewer than* $2^{1+(1-1/(8N))P}$ *inputs both transpose by* T *and satisfy every episode bound* $|D_h(I)|\le m$. *There is a fixed* $\delta>0$ *for which one may take* $s\le\delta P$, $m=\lfloor\delta P\rfloor$. *Outside that exception, some secondary episode has at least* $\lfloor\delta P\rfloor+1$ *distinct actual modifications, more than* δP *successful steps, and work-head span at least* $\lfloor\delta P\rfloor$.

Proof (not formalized). Write $\lambda_M=\lceil\log_2 K_M\rceil$ and $L_0=\lceil\log_2(2k(T+1)\ell)\rceil$. Since $K_s\le K_M$, the compiler's bit bill is at most

$$q_{\mathrm{src}}+q_{\mathrm{obs}}+A_0\lambda_M+k(\ell+1)L_0+\lceil\log_2 B_{\mathrm{cache}}\rceil+2.$$

The last three terms are $O(\log(P+1))$, so the supplied reserves imply $q_*\le P/(64N^2)$ and $k\ell r\le P/(128N^2)$ uniformly at large d. Together with Z_0 this gives the label bound of Lemma D.1. Apply it and retain the half-set factor. For the quantitative choice it suffices to take $\delta\le 1/6$ and

$$3\delta\log_2\frac{e\lceil 4C+1\rceil(2^a-1)}{3\delta}\le\frac{1}{512A_0N^2}.$$

Eventually $M \le 3\delta P$ and the sparse binomial bound, with its ceiling, fits the reserve. Such $\delta > 0$ exists. Negation of the episode promise gives the distinct-address count. One successful step changes at most one address on this tape, and j distinct integer addresses have span at least $j-1$, proving the remaining claims. The checked `ExecutionPrefix.many_episode_changes_duration_span` counts actual changed addresses only before halt. It proves the strict duration bound and the span bound even when the declared episode contains the halting configuration; that configuration contributes no successful write. □

Throughout the episode its tape and the current carrier are two different unmatched tapes; when $k > 2$ the other identity may vary. This is not an independence or output-usefulness conclusion. The shared static scan seed conditions from Appendix D carry over without a seed per episode. For static reference families the cache reduces to a larger-radius case; for public or prefix-only historical bases, a polynomial catalogue enlargement already suffices. The added interface here permits raw-dependent historical bases while current masks and cutoffs evolve.

F Persistent marginal information

All logarithms are base two. For a finite random variable W, $H_\infty(W) = -\log_2 \max_w \Pr(W = w)$. A worst-case conditional bound ranges over attained conditioning values of positive probability. It is stronger than a bound on the average guessing probability. Neither notion implies independence of two work words.

Lemma F.1 (One subset for every context)**.** Lean: `context_subset`, `context_half_mass`, `context_conditional_bound` [`ContextPruning.lean`]

Let (Ω, μ) be a finite probability space, $\mu(B) = p > 0$, and let $Z_1, \dots, Z_\ell$ be context variables. For $\theta > 0$ there is $D \subseteq B$ with $\mu(D) \ge p - \ell\theta$ and

$$\mu(D \cap \{Z_i = z\}) \ge \theta \mu(Z_i = z)$$

for every context attained on D. If, for every i, v, w, z, $\mu(B, W_{v,i} = w, Z_i = z) \le c\mu(Z_i = z)$, with $c \ge 0$ and $\ell \ge 1$, choosing $\theta = p/(2\ell)$ gives on this one D of mass at least $p/2$

$$\Pr_D(W_{v,i} = w \mid Z_i = z) \le 2c\ell/p.$$

Outline of the formal proof. For $E \subseteq B$, let $S_i(E)$ be the union in Ω of all i-context fibres meeting E. Maximize the finite potential $\Phi(E) = \mu(E) - \theta \sum_i \mu(S_i(E))$. For a maximizer D, $\mu(D) \ge \Phi(D) \ge \Phi(B) \ge p - \ell\theta$. If an attained fibre $F = D \cap \{Z_i = z\}$ violated the asserted lower bound, removing F would lose $\mu(F)$ but remove the whole penalty $\theta\mu(Z_i = z)$ for this context. All other penalties can only decrease, contradicting maximality. Finally divide the joint envelope by this fibre lower bound. The global conditioning factor $\mu(D)$ cancels. There is no extra factor for the number of word variables sharing each context partition. □

Theorem F.2 (Persistent information beyond the strict output prefix)**.** Lean (partial): `ConditionalPersistence.of_small_thin_source` [`PersistenceSource.lean`]
Not formalized: the native catalogue counting theorem. Given its exact thin-source cardinality bound, including its factor two, Lean constructs the original uniform law, escaping set, common live interval, retained set, conditional whole-word bound and uniform threshold.

Fix $k \ge 2, a, C$ and the constants ρ, η of Theorem B.2. There are constants $\alpha, \beta, p_0 > 0$ such that, for every sufficiently large d, every $b \ge 1$ and machine with $q \le \rho P/2$, the following holds. If S is a

timely correct set of size at least $2^{(1-\eta/2)P}$, there are two fixed tapes, one fixed interval of at least $\beta P/4$ live configuration times, and one $D \subseteq S$ with $|D|/|S| \geq p_0/2$, such that at every time of the interval both whole work words have worst-case min-entropy at least $\alpha P/2$ conditional on the strict output prefix, under the uniform law on D. On every selected input, at least αP output bits remain undelivered throughout.

Proof (not formalized). Set $h_0 = \rho/[4(k+1)]$, $A_0 = 2^a$, $\alpha = h_0/4$, $\beta = h_0/[4\log_2(6A_0)]$, $R = \lfloor \beta P \rfloor$, $\ell = \lfloor R/2 \rfloor$ and $K_0 = \lceil 2^{\alpha P} \rceil$. Fix the original uniform law μ on S. A one-tape vertex is its entire word on $[-T_*, T_*]$ and its head position. Writing one symbol and making one unit move gives at most $3A_0$ outgoing and $3A_0$ incoming neighbours. Hence an undirected radius-R ball has at most $(6A_0)^{R+1}$ vertices.

For each time t, prefix z and tape, take its K_0 most probable positive conditional vertices, or all if fewer exist. Outside this set each conditional atom has mass at most $1/(K_0+1)$. Define a public catalogue at (t, y) by thickening the top sets at $u = (t-l)_+$ and prefixes $y[0 : (|y| - r)_+)$ for $0 \leq l, r \leq R$. Use radius-R balls and map each vertex to its whole word, blank outside $[-T_*, T_*]$. This square overcount includes every actual historical prefix: at lag l, the prefix loses at most l bits. Forgetting the head cannot increase the catalogue size, so its size is at most

$$K = (R+1)^2 K_0 (6A_0)^{R+1}.$$

Every actual historical prefix occurs among these candidates because at most one bit is appended per transition. Its logarithm, including the ceiling, is at most $(h_0/2)P + O(\log P) \leq h_0 P$ eventually. The catalogues are fixed from the original law before applying Theorem B.2.

That theorem implies that more than half of S have a reached time t and two words escaping their current catalogues. Every vertex of both tapes in the preceding R steps then avoids its corresponding conditional top set. The witness time exceeds R: otherwise the certain initial blank vertex and its actual path put every word in its catalogue. Choose $I_j = \{j\ell, \ldots, (j+1)\ell - 1\}$ with $j = \lfloor t/\ell \rfloor - 1$. It lies in $[t-R, t)$ and is strictly before halt, even if t itself is the halting configuration. There are at most $M_0 = \binom{k}{2}(\lceil 8C/\beta \rceil + 1)$ pair/grid choices. One cell B of this partition has mass $p \geq p_0 = 1/(2M_0)$.

At each time in this fixed interval, both tapes avoid their conditional top sets throughout B. Forgetting the head position costs at most $n = 2T_* + 1$ atoms, giving the original-law envelope

$$\mu(B, W_{v,t} = w, Z_t = z) \leq \frac{n}{K_0 + 1} \mu(Z_t = z).$$

Apply Lemma F.1 to the ℓ time-prefix partitions. One D of mass at least $p_0/2$ has every conditional word atom at most $2n\ell/[(K_0+1)p_0]$. Its negative logarithm is $\alpha P - O(\log P) \geq \alpha P/2$. A positive attained vertex outside the top set also witnesses at least $K_0 + 1$ inputs in its original time/prefix fibre. Correctness fixes $|z|$ distinct raw bits, so that actual fibre has size at most $2^{P-|z|}$. Thus $P - |z| \geq \log_2(K_0+1) \geq \alpha P$. Adaptive output timing can further shrink the fibre; equality with $2^{P-|z|}$ is not assumed. All bounds are uniform for $P \geq d^2$.

The formal endpoint retains the native catalogue bound $2^{1+(1-\eta)P}$ as its sole unproved input. The source-size bound and $\eta P/2 \geq 2$ leave at least half the original uniform source outside that thin set. The shared `LeanMachine.FiniteSourceEscape.half_mass_pair_escape` constructs that event and its actual two-tape witnesses. The remaining selection, probability and rate estimates are formalized, with the dimension threshold preceding the width, machine and source quantifiers. `ConditionalPersistence.of_half_mass_escapes` also works under any strictly positive finite source law satisfying the literal escape condition. The context-pruning step uses one retained set for every selected time and tape. The prefix-fibre bound follows from actual native correctness; no full-size fibre, independent tape words or entropy lower bound is assumed. □

Taking no prefix condition in the same argument gives the ordinary persistent-marginal theorem. The stronger conditional result still permits the two words to be identical. It cannot be summed over times as repeated fresh information.

F.1 Future successful reads and remote returns

Define the future-readable word at source time t by retaining the current symbol exactly at cells visited in successful steps $u \geq t$ before halt; at and after halt it is blank. Call it $L_h(t)$. The unused halt-test tuple does not count as a future read. Its nonempty future address set is an integer interval, by unit motion.

Theorem F.3 (Source future-readable catalogue bound)**.** Not formalized; a complete proof is given below.

For fixed k, a, C there are $h, e, \rho' > 0$ such that, uniformly in b at large d, every public time-indexed family of catalogues of size at most K with $\lceil \log_2 K \rceil \leq hP$ has fewer than $2^{(1-e)P}$ timely correct inputs for which at every source time at most one $L_h(t)$ lies outside its catalogue. One may allow $q \leq \rho' P$. No online access to future information is given to the source.

Proof (not formalized). Use the aligned bit simulator of [4] with block size $s \geq 1$ and physical batch length $B = 10s + 1$. Its literal tape word at time Bj equals the source word at time js. Each batch reads/writes a public $3s$-cell window around its anchor; outside that window the word remains the source boundary word throughout the batch. Its positions at successful time u differ by at most $2s$ from the source at $\min(s\lfloor u/B \rfloor, t_X - 1)$, including the final batch and stationary drain. These identities follow from the simulator's boundary invariant and its actual sweep/carry rule, with source offset in $[0, s-1]$ and physical offset in $[-s, 2s-1]$. The possible terminal replacement $t_X \mapsto t_X - 1$ costs one unit.

Fix one full simulator trace of successful horizon U. Hardwire its last successful visit flags and clock in a native eraser: simulate each old tuple, but replace a work write by blank at the last visit to that cell. Force halt at U, because the erased final tuple need not itself trigger the source halt rule. On this trace fibre, induction shows its word is exactly the simulator word masked to future successful visits, with unchanged output and transition count. It has at most $1 + Q_{\mathrm{sim}}(U+1)$ states.

For an ordinary batch $j < \lfloor t_X/s \rfloor$, let J be this trace's public future simulator interval and I_X the source future interval at js. The shadow gives $J \subseteq I_X + [-2s, 2s]$. Remove the first and last $2s$ cells of J. The remainder is in I_X. Outside these at most $4s$ cells and the current $3s$ write window, the eraser word is exactly the source future-readable word masked to J. Thus a source catalogue induces one of size at most KA^{7s}. All heads use the same source time js. In the final batch and drain the entire future word is supported in the public $3s$ window; at halt it is blank. This covers divisible source runtimes and $s = 1$. No source future-interval endpoints are supplied as free metadata.

Set $C_N = 11C + 1$ and first choose the catalogue constants of Theorem B.2 for (k, a, C_N), independently of s. Write them ρ_N, η_N and $h_N = \rho_N/[4(k+1)]$. The published successful-input simulator bound gives at most $2^{a_0 P/s}$ full traces, where $a_0 = (2C+1)\log_2(1 + 2 \cdot 3^{k+1})$, for large P. Choose fixed s with $a_0/s \leq \eta_N/4$. Its control cardinality is at most QD_s, with

$$D_s = (11s+2)s^{k+2}3^{3s+k+1}(2^a)^{3sk}2^{2s+2}.$$

This counts the actual phase, offsets, raw/work windows, carry directions, pending/scheduled bits and stopped flag. The eraser consequently costs only $q + \lceil \log_2 D_s \rceil + \lceil \log_2(U+1) \rceil + 1$ bits. Take $\rho' = \rho_N/4$ and $h = h_N/2$; fixed and logarithmic overhead eventually fit the remaining reserves.

Apply the ordinary time-indexed catalogue bound separately on every trace fibre and sum, retaining its factor two. The total is less than $2^{1+(1-\eta_N+a_0/s)P} < 2^{(1-\eta_N/2)P}$ eventually. Thus $e = \eta_N/2$ works. No largest-fibre selection loses the original set. □

Corollary F.4 (Persistent information on original remote returns)**.** Lean (partial): `PersistenceFutureSource.of_thin_source_bound` [`PersistenceFutureSource.lean`]
Not formalized: the source future-readable catalogue counting theorem. From its explicit thin-source bound, Lean derives the original uniform law, one common event and live interval, whole-word and remote-word atom bounds, actual successful returns and a threshold uniform over positive widths.

For fixed $k \geq 2$ and resource-dependent $\alpha, \beta, p_0 > 0$, every sufficiently large timely correct set S of size at least $2^{(1-e/2)P}$ contains a subset of relative size at least p_0, two fixed tapes and a common interval of at least βP live times on which both future-readable words have min-entropy at least αP. The same holds, with smaller constants, after blanking a fixed radius-$\alpha' P$ window about each current head. Every nonblank cell of the latter word lies on an actual macroscopic return straddling that checkpoint. These are marginal, not prefix-conditioned, claims.

Proof (not formalized). The pair $(L_h(t), y_h(t))$ follows the same bounded-degree one-tape graph used above: only the old scanned cell changes, retaining its write if revisited later and otherwise becoming blank. Thicken the K_0 most likely vertices at all nearby times under the original uniform law on S, now without prefix indices. At radius $R = \Theta(P)$ the catalogue size is $(2R+1)K_0(6A_0)^{R+1}$, which fits Theorem F.3 by choosing small exponential rate and radius. Escape on more than half of S gives a fixed pair/grid interval of constant relative mass, exactly as in the preceding proof. Full vertex atoms there are at most $1/(K_0 p_0)$; forgetting the head costs at most $\log_2(2T_* + 1)$ bits.

Masking a radius-H window has fibres of size at most $(2T_* + 1)A_0^{2H+1}$ on full vertices. Choose a sufficiently small fixed $H/P > 0$; linear min-entropy survives. A nonblank retained cell was written at some previous visit and has a future successful visit. Its last past and next future visits are consecutive and its distance from the current head exceeds H, so the checkpoint witnesses the claimed return. All selected times are before halt. Neither this graph argument nor the eraser gives the original source a future-read oracle or proves that its later read matters semantically to the output.

The formal proof uses one finite observation window $[-T_*, T_*]$ for the whole source law, while retaining each input's own first halt time in its future-readable word. Frozen steps after that halt are never future reads. `PersistenceFutureRates.of_half_mass_escapes_all_widths` proves the selection and quantitative bounds under arbitrary finite laws; the stated uniform-source result derives its half-mass escape event from the thin-source count. The native eraser and source-catalogue theorem remain the only unformalized dependency of this consequence. □

G Joint information and exact cut records

The following consequences make explicit what is counted jointly. They do not justify adding the information of a reused record across times, rectangles or scales. All entropies in this section are in bits.

Lemma G.1 (Finite joint invoices)**.** Lean: `joint_record_entropy_identity`, `joint_record_information_lower`, `joint_record_card_bound` [`InformationAccounting.lean`]
On any finite probability space, if $x \mapsto (M(x), E(x), R(x), Z(x))$ is injective, then

$$H(Z \mid M, E, R) = H(x) - H(M) - H(E \mid M) - H(R \mid M, E).$$

Upper bounds on the last two conditional terms give the corresponding lower bound on the import term. Separately, an injective map from a finite set S to an e-bit exterior and a record C satisfies $|S| \le 2^e |\operatorname{range}_S C|$; only the attained joint range is paid.

Outline of the formal proof. An injective deterministic observation has the entropy of its source. Expand conditional entropy and telescope the three nested pair entropies. The imported library uses natural logarithms; division by $\log 2$ gives the displayed bit convention. For cardinality, pair the exterior with the record viewed in the finite type of its attained image, and apply cardinality monotonicity to this injection. Constructing a native record with that injectivity property remains a separate obligation. The metadata-dependent support form is also checked in `conditional_entropy_le_mean_log_support` and `joint_import_information_from_support`: a menu of possible record values at each metadata value costs the expected logarithm of that menu's cardinality. Empty and zero-mass metadata fibres contribute zero. This proves the variable-length entropy step without assuming every metadata fibre has positive mass. □

Corollary G.2 (Conditional information in the import record)**.** Lean: `NativeReplayClass.code_injective`, `NativeReplayFamily.native_import_information_saving` [`NativeImportInformation.lean`]

Give a nonempty timely correct set S its uniform law and make the choices of Theorem 3.3 deterministically. Let Z_0 be its geometry metadata, E the exterior raw word, R_0 the ordinary state/window/slab records, and Z the single tagged import record. Put $D = P - \log_2 |S|$. Then

$$\begin{aligned} H(Z \mid Z_0, E, R_0) &\ge \mathbb{E}|U| - D - H(Z_0) - H(R_0 \mid Z_0, E) \\ &\ge \mathbb{E}[|U| - b_0] - D - H(Z_0), \\ b_0 &= a\mathbf{1}_{a_0>0} W + a(k-1)r(2H+1) + rq. \end{aligned}$$

Here r, U, a_0 are the selected input-dependent geometric quantities. More generally the same two bounds hold under any one probability law on S, with $D = P - H(X)$.

Outline of the formal proof. For each metadata value the checked class contains actual execution prefixes, entry observations, extrema, output counters and the computed import record. The canonical run decomposition and counter intervals give the replay hypotheses. The native replay theorem therefore proves the record injection; injectivity is not a field assumed by the class interface.

Apply the finite entropy identity to the metadata-indexed family of these actual records. Conditional on Z_0, the exterior has a fixed coordinate set and at most $P - |U|$ bits, so $H(E \mid Z_0) \le P - \mathbb{E}|U|$. Rearranging gives the first inequality. At each fixed metadata value the ordinary record has support at most 2^{b_0}, yielding the second after conditioning on E and averaging. Finite dependent record types are tagged by the metadata already in the conditioning variables; they incur no extra choice cost. All terms use the same law, and Z includes tags as well as values. The formal statement uses natural logarithms and permits arbitrary finite free cell sets and demand maps. Dividing by $\log 2$, specializing to the slab, and using $H(X) = \log_2 |S|$ for the uniform law gives the displayed version. □

Proposition G.3 (Exact frozen-trace cut invoice)**.** Lean: `native_cut_recovers`, `native_frozen_full_invoice`, `native_frozen_field_invoice`, `native_frozen_cut_entropy_lower` [`FrozenCutInvoice.lean`]

Let a nonempty correct set S share one complete head/append trace through the relevant halt horizon. Let Q_0 be D_0 distinct demanded raw coordinates, R their designated correct append roots, and let Γ

be an endpoint-inclusive cut from R to every actual read of Q_0. If C_Γ is the joint full step-value record at the cut, then

$$D_0 - (P - \log_2 |S|) \le \log_2 |\operatorname{range}_S C_\Gamma| \le (q + ka + 2)|\Gamma|.$$

If only post-state and written work symbols are retained at every cut step, together with the emitted bit at roots in Γ, the sharper field bill is

$$D_0 - (P - \log_2 |S|) \le (q + ka)|\Gamma| + |\Gamma \cap R|.$$

The statement holds for any fixed output permutation and any entry width. Moreover, under any one probability law on S, the same joint full record satisfies

$$H(C_\Gamma \mid X|_{[0,P)\setminus Q_0}) \ge H(X) - (P - D_0).$$

Outline of the formal proof. Give the decoder X outside Q_0 and the cut record. A cut root's bit is in its full value. For any uncut root, follow all its uncut ancestors in the fixed dependency graph. Every such vertex reads outside Q_0, since otherwise its ancestor path contradicts separation. Reconstruct these vertices in increasing time. Each earlier state/work parent is either retained and already reconstructed, cut and supplied, or initial and known. The complete common old tuple determines the instruction and output. Latest work visits, and all physical positions and append ordinals, are fixed by the trace. Hence all demanded output bits, and therefore X on Q_0, are recovered. This proves an injection into the exterior bits and the *joint range* of the cut record, giving the first bound. Its product alphabet gives the second.

For the field version, retained descendants need only a cut predecessor's post-state and work writes. They never read its output. Only a cut desired root needs its own emitted bit; its append flag is already fixed by the trace. Count those fields to obtain the sharper formula. Its root term cannot be discarded: a cut root can emit its raw input bit while keeping all state and work fields constant. A single encoder pooling different privately selected traces or cuts across inputs must charge those choices; this formula does not remove the geometry menu from Theorem 5.3. Applying the universal frozen-fibre inequality separately to each fixed cut needs no additional cut-choice bill, as in Proposition H.4.

The checked statement uses arbitrary input length and designated roots whose actual native output bits equal the demanded coordinates. It proves the reconstruction from the actual field-level transition recursion and the literal endpoint-inclusive dependency-path separator. Its trace agreement is required only through the horizon, so the complete common trace in the proposition is sufficient. Neither injectivity nor cone determination is an additional premise. For the arbitrary-law statement, the proved exterior/record injection identifies their joint entropy with $H(X)$. Subtract the exterior's entropy, which is at most $P - D_0$ bits. This uses the native reconstruction just proved, not a further hypothesis of injectivity. □

H Transporting certificates to original executions

The direct replay arguments cover every positive width, but they do not subsume two other conclusions: expansion from *every* large output set to the original crossing set, and storage capacity on an original run's entry quotient. This appendix retains those conclusions. Their finite interfaces use the actual aligned simulator of [4]; none identifies its dependency graph with the original graph.

H.1 A guarded projection

Fix a halting source execution with $t \geq 1$ transitions and an integer $s \geq 1$. The aligned simulator has k corresponding work heads and halts after

$$B_0 = 10s + 1, \qquad U = B_0(\lfloor t/s \rfloor + 1) + s$$

transitions. The extra complete terminal batch is present even when $s \mid t$. Its final s steps form a stationary drain. Define

$$q(u) = \min\big((\lfloor u/B_0 \rfloor + 1)s, t - 1\big), \qquad 0 \leq u < U.$$

Write x_h for source positions and y_h for physical simulator positions. At batch j the common anchor is $s\lfloor x_h(js)/s \rfloor$, and both the physical batch and the corresponding closed source interval lie in the anchor window $[-s, 2s - 1]$. At the terminal plateau the preceding source step adds at most one unit. These are consequences of the actual simulation invariant in [4], not of an idealized rescaled schedule.

Lemma H.1 (Guarded path projection)**.** Lean: `TransferGuards.native_guarded_projection` [`TransferNativeProjection.lean`]

For every source vertex set B, there is a simulator set E with

$$|E| \leq [4B_0 + k(6s + 1)]|B|$$

such that every surviving simulator dependency path projects to a source path with endpoints $q(u), q(v)$ avoiding B, endpoints included. Moreover $|y_h(u) - x_h(q(u))| \leq 3s$ for every corresponding head, including the input head.

Outline of the formal proof. The checked execution invariant of the actual compiler proves the position bound, including the terminal batch and drain. Any two time-ordered source visits in a closed band admit a backward dependency path within that band, even if the source walk leaves both sides between them. Follow the state edge when its predecessor stays in the band; otherwise jump to the actual last prior visit to the current boundary. Unit speed supplies that visit. Time decreases at each step. Thus endpoint shadow alone projects a simulator work hop at cell z to a source path in $[z - 3s, z + 3s]$. Equal projected endpoints require only a singleton path.

Mark every whole physical batch whose closed projected source interval meets B. A source cut time belongs to at most four such batches, including the extra terminal batch and drain. Each costs at most B_0. Also mark the later endpoint of a simulator work hop whenever its projected interval strictly straddles a cut time b and its cell is within $3s$ of $x_h(b)$. For each (b, h, z) at most one consecutive-visit interval straddles b: such intervals have disjoint interiors and q is nondecreasing. This costs at most $k(6s + 1)|B|$. These are the sets whose union is E.

A surviving state edge projects either to a singleton or to a source spine interval disjoint from B. A surviving work edge has the band path just constructed. A cut vertex on that path would either be a projected endpoint, contradicting the whole-batch guard, or lie strictly between them, contradicting the hop guard. Concatenation proves the path statement. This uses every actual work last-visit edge and does not introduce input-revisit edges. □

Let α_i, u_i be the actual source and simulator append times for output ordinal i. Ordinals are zero-based and matched even when output bit values repeat. Let $O(v)$ count source appends strictly before time v.

Lemma H.2 (Output look-ahead and cones)**.** Lean: `TransferCones.native_cone_transport_interior`, `TransferCones.native_cone_transport` [`TransferNativeCones.lean`]

For every existing ordinal, $\alpha_i \le q(u_i)$ and $O(q(u_i)+1) \le i+2s$. Put

$$G_s = 4B_0 + k(6s+1) + 2s + 1.$$

For every B there is a guard Γ of size at most $G_s|B| + 2s$. Every append root $u_i \notin \Gamma$ has $i+2s$ defined, and its root-exempt input cone satisfies

$$\mathrm{Cone}_{N_s}(\Gamma, i) \subseteq \mathrm{Dil}_{3s}\big(\mathrm{Cone}_M(B, i+2s)\big) \cup \mathrm{Boundary}_{3s}(P).$$

The dilation is in valid raw coordinates; the boundary comprises the first and last $3s$ positions, with overlap allowed.

Outline of the formal proof. At physical batch start j, the emitted output count is $E_j = s\lfloor O(js)/s \rfloor$ and fewer than s source outputs are pending. The compiler emits a prefix of length either zero or s and retains the suffix. Thus its FIFO order gives $\alpha_i \le q(u_i)$. Through the source step $q(u_i)$, inclusive, at most $s+1$ new outputs follow $O(js)$. For a batch emission $i \ge E_j$, this gives $O(q(u_i)+1) \le E_j + (s-1) + (s+1) \le i + 2s$. The drain has fewer than s residual outputs and satisfies the same bound. These actual append-order facts are checked in `TransferFIFO.actual_sourceRoot_le_clock` and `TransferFIFO.actual_ordinal_lookahead`. Hence $q(u_i) < \alpha_{i+2s}$ when that future ordinal exists.

Take the guard of Lemma H.1. Add the last at most $2s$ append roots and every root u_i whose closed interval $[\alpha_i, \alpha_{i+2s}]$ meets B. A cut time belongs to at most $2s+1$ these intervals: their smallest and largest indices differ by at most $2s$, since append times increase strictly. This proves the guard bill. For a surviving root, prepend the B-free source spine from α_{i+2s} to $q(u_i)$ to the projected path. At a raw endpoint, the input-head shadow places the simulator raw address within $3s$ of the source address. If that source address is valid it belongs to the source cone. Otherwise the original valid simulator address is in the displayed boundary. Singleton paths and endpoints in cuts are included. The look-ahead bounds output counts, not the physical waiting time of a pending output bit. The formalization also gives the version before adding the terminal roots: the guard costs at most $G_s|B|$, and the inclusion holds whenever the future ordinal exists. This is the version used for interior entry lanes. Every constructed guard consists of actual simulator vertices. □

H.2 Finite raw-input cuts and erosion

For $U \subseteq [0, P) \cap \mathbb{Z}$, regard U as a subset of the entire integer line and define

$$E_s(U) = \{z \in \mathbb{Z} : [z-3s, z+3s] \cap \mathbb{Z} \subseteq U\}, \qquad C_s = 4B_0 + k(6s+1) + 2s - 1.$$

Thus erosion pays the physical input boundaries as well as internal gaps.

Lemma H.3 (Finite interval erosion)**.** Lean: `interval_union_erosion_loss`, `demanded_erosion_loss` [`InputErosion.lean`]

If a finite integer set U is a union of r integer intervals, then $|U \setminus E_s(U)| \le 6sr$. Consequently, for every $V \subseteq U$, $|V| \le |V \cap E_s(U)| + 6sr$. The intervals need not be disjoint.

Outline of the formal proof. Every point of an interval outside its two end strips of width $3s$ has its whole radius-$3s$ neighborhood in that interval, hence in U. Therefore $U \setminus E_s(U)$ lies in the union of the $2r$ end strips, whose total cardinality is at most $6sr$. The second inequality partitions V into its eroded and noneroded parts. The Lean statement allows an arbitrary nonnegative integer radius and a finite interval cover. □

Proposition H.4 (Finite eroded-input transfer)**.** Lean:
`TransferEroded.native_eroded_fibre_invoice`,
`TransferEroded.native_eroded_fibre_invoice_cover` [`TransferErodedInvoice.lean`]

On an actual source run and its aligned simulator, a source cut B from any set of existing append ordinals to all source reads of U lifts to a simulator cut Γ from the matched append roots to all simulator reads of $E_s(U)$, with

$$|\Gamma| \le C_s|B| + 2s - 1.$$

This finite geometric assertion needs no correctness or common source trace. On a nonempty common simulator trace fibre S of correctly transposed inputs, put $D_S = P - \log_2|S|$ and let v_s be the simulator's complete step-value width, including its output field. For every $X \in S$, every $V \subseteq U \subseteq \{0,\dots,P-1\}$ and every original cut B separating the demanded roots for V from all original reads of U,

$$|V \cap E_s(U)| \le D_S + v_s(C_s|B| + 2s - 1).$$

If U is a union of r integer intervals, then $|U \setminus E_s(U)| \le 6sr$; in particular the left side can be replaced by $|V| - 6sr$. These finite assertions hold at every positive entry width.

Outline of the formal proof. The physical input head satisfies the same shadow estimate $|y_0(u) - x_0(q(u))| \le 3s$ used in Lemma H.1. In an ordinary core both positions lie in the aligned anchor plus $[-s, 2s-1]$; the terminal replacement by the last live source step costs at most one unit, and the stationary drain has error at most s. These estimates apply to the actual input-head coordinate, including the terminal core. Therefore a simulator read at a point of $E_s(U)$ projects to an actual source read in U. Its connector is the singleton source vertex $q(u)$; no input-revisit edge is introduced.

Use the edge guards of Lemma H.1, costing at most $[4B_0 + k(6s+1)]|B|$. In the source graph with B deleted, let R_U be the vertices reaching a read of U. Its intersection with a B-free state-spine interval is an upper terminal segment, so there are at most $|B|+1$ positive activations. At an activation time c, the FIFO invariant in Lemma H.2 leaves at most $2s-1$ output ordinals whose actual source root precedes c but whose simulator shadow is at least c. Delete those matched simulator roots. This costs $(2s-1)(|B|+1)$. Every surviving simulator path to an eroded read projects to a source path avoiding B and ending at a read of U. Its projected root lies in R_U, whereas its actual source root lies outside R_U by separation; the activation guard forbids this. Singleton paths and cut roots are covered by the endpoint-inclusive guards. If the source has no successful step there are no append ordinals and the assertion is vacuous. The complete geometric lift, including this zero-time case, is `TransferEroded.native_eroded_cut_lift`.

Restrict the demanded source coordinates to $V \cap E_s(U)$ and apply the lift. Its resulting separator is on the one simulator trace shared by S. The frozen-trace injection of Proposition G.3 gives $|V \cap E_s(U)| \le D_S + v_s|\Gamma|$, proving the displayed bound. The weaker coefficient v_s+1 is an immediate consequence; a full step record already includes the output field and needs no further root surcharge. This direct injection also explains the all-width finite claim; no canonical-width asymptotic theorem is used. The cut can depend on X, U, V because the frozen-trace assertion holds for every cut on that fixed fibre, with no restriction to inputs sharing the original trace.

Apply Lemma H.3 to the given interval cover. The maximal interval components are one such cover. Highly fragmented sets may have empty erosion; this cost is not discarded. □

H.3 An original entry quotient carries a large storage code

For one reference source run X and cut B, let $Q_X^B(e)$ contain entry f if any bit of f belongs to the source root-exempt cone of any of the b output bits of entry e. This is the full quotient: all lanes

and self-edges are retained. For a relation Q on J, let $\mathrm{Cap}_m(Q)$ be the base-two logarithm of the largest storage-code cardinality over m-bit symbols. Recovery means that agreement on all helpers in $Q(e)$ forces agreement at e. Write $\Delta_m(Q) = m|J| - \mathrm{Cap}_m(Q)$; both are zero for $m = 0$.

Theorem H.5 (Theorem Q: capacity of original entry relations)**.** Lean: `TransferEntry.native_original_entry_capacity` [`TransferEntryNative.lean`]

Let S be a nonempty set of correctly transposed inputs with one common complete simulator trace. Set $m = (b - 6s)_+$, $D_S = P - \log_2 |S|$, and let v_s be the complete step-value width of the simulator. For every reference $X \in S$, every source vertex set B on that run and every entry set J,

$$\Delta_m(Q_X^B[J]) \le D_S + v_s G_s |B|.$$

The extracted code need not contain X, and its members need not have the same original trace. The finite statement holds at every $b \ge 1$.

Outline of the formal proof. If $m = 0$ the conclusion is immediate. Otherwise use the guard in Lemma H.2 without the final-output tail, of size at most $G_s|B|$. An interior lane $3s \le \gamma < b - 3s$ is never in that tail. Partition S by the joint complete guard values and every raw bit outside the interior lanes of J. There are at most $2^{v_s G_s |B| + P - m|J|}$ labels. A largest cell, projected to the remaining lanes, is an injective image and has logarithmic cardinality at least $m|J| - D_S - v_s G_s |B|$.

For a guarded interior root its output bit is fixed by the complete guard value. For another root, the output look-ahead stays within entry e. If an interior raw helper survives the fixed exterior, its distance-$3s$ source neighbor stays within the same helper entry. Thus every nonconstant helper is in $Q_X^B(e) \cap J$. The fixed-trace cone-determination theorem of [3] gives recovery of each bit throughout this cell. The checked cone-determination step is `nativeInputCone_determines`; guarded roots use the output field already present in the complete record. Boundary helpers are fixed. The reference run fixes B, Q_X^B and the simulator guard, whereas only the simulator trace is required to be common. This proves the storage-code property and the capacity inequality. □

Corollary H.6 (One event for all original capacity certificates)**.** Lean: `TransferEntry.uniform_entry_success_tail`, `TransferEntry.certified_original_cut_lower_bound` [`TransferEntryNative.lean`]

Fix a machine, a linear time constant, canonical width $b(d)$, and $0 < \delta < 1$. Some fixed s and all sufficiently large d have the following property: fewer than $2^{(1-\delta/2)P}$ timely correct inputs violate any of

$$\Delta_{b-6s}(Q_X^B[J]) \le \delta P + v_s G_s |B|, \qquad \text{for all } B, J.$$

If $0 < \alpha, \rho \le 1$, $|J| \ge \alpha d^2$ and $\mathrm{Cap}_{b-6s}(Q_X^B[J]) \le (1-\rho)(b-6s)|J|$, choosing $\delta = \rho\alpha/4$ gives, on the good event at large d, $|B| \ge \rho b |J|/(2 v_s G_s)$.

Outline of the formal proof. The successful-subset simulator resource theorem of [4] bounds the number of attained simulator traces by $2^{aP/s}$ for a fixed a. Choose s with $a/s \le \delta/2$ before dimension grows. Fibres of size at least $2^{(1-\delta)P}$ satisfy every finite inequality above. The total number of successful inputs in smaller attained fibres is strictly below $2^{(1-\delta/2)P}$. There is no union bound over B, J or reference inputs. For the final assertion take $b \ge 24s$: the deficit is at least $3\rho b|J|/4$, whereas $\delta P \le \rho b|J|/4$, and rearrange. □

Acyclic entry relations have singleton storage codes and are a special case. Bit-level acyclicity is insufficient: shifting a target by $2s$ can introduce self-dependence within an entry. No separation, degree or missing-self-edge hypothesis is supplied automatically. A capacity theorem is useful here only after its own premises have been established on the actual quotient.

H.4 Expansion from every large output set to original crossings

Let CL_h be the earlier endpoints of same-tape crossing consecutive revisits whose two excursions exceed $2h$, with the strict crossing order of [3].

Theorem H.7 (Theorem E: original adaptive crossing expansion)**.** Lean: `TransferCrossing.uniform_original_crossing_expansion` [`TransferCrossingTail.lean`]

Fix a native machine, $C \geq 0$ and $\alpha > 0$. At canonical width there are $\eta, \lambda, \beta > 0$ such that, at every sufficiently large d, fewer than $2^{(1-\eta)P}$ timely correct inputs fail the following simultaneous statement about their original execution: for every $|V| \geq \alpha P$, every vertex cut from the actual append roots $\mathrm{Out}_M(V)$ to $\mathrm{CL}_{\lfloor \lambda P \rfloor}$ has size at least $\beta|V|$. No original-trace heaviness assumption is imposed.

Outline of the formal proof. We first give the finite crossing-target addition to Lemma H.1. Let $H \geq h \geq 1$, and let K_0 bound the number of grid-H epochs of each simulator head. Each epoch has diameter less than $2H$. A crossing pair $p < p' < u < u'$ has opposite polarities after one choice of sign: its first hop is above base a, and its second below base $b > a$. Choose an epoch containing a maximum of the first hop and one containing a minimum of the second. Their fixed signed extremizers t_1, t_2 satisfy $p < t_1 \leq p' < u \leq t_2 < u'$. Let t_0 minimize the signed walk before t_1, and t_3 maximize it after t_2. The resulting zigzag has overlap

$$[L_0, U_0] = [\max(y(t_0), y(t_2)), \min(y(t_1), y(t_3))]$$

containing both bases. For fixed sign, ordered epoch pair and endpoint role, each base level determines at most one endpoint: distinct consecutive-visit hop interiors at one level cannot both contain the fixed extremizer.

Put $R = 3s$ and $w = 2h + 2R + 2$. Discard targets whose assigned level is within w of either end of the overlap. The preceding injection bounds their number by $4kK_0^2(w+1)$. Indeed the two extremizer epochs are distinct: their values differ by more than $2H$, whereas one epoch has diameter less than $2H$. The checked label records the sign, endpoint role, two epoch indices and a level offset at most w. Reversing the strictly ordered epoch pair encodes the upper end strip, so no extra strip bit is needed. The label space has size $4K_0^2(w+1)$ per tape, and actual consecutive-visit uniqueness makes the encoding injective. At a remaining first-role target of level a, take $\ell = a - R - 1$ and $c = \ell + 2h + 1$. The shadowed zigzag visits below ℓ, above c, below ℓ, then above c. The last ℓ visit before $q(v)$ and first after it form a genuine upper hop; the neighboring c visits supply a crossing lower hop. Both excursions exceed $2h$. For a second-role target at b, reflect the construction with $\ell = b + R + 1$, $c = \ell - 2h - 1$; the crossing partner occurs *before* the chosen lower hop. In either case the checked band-path argument connects $q(v)$ backward to the appropriate source endpoint within the common band $[y(v) - R - 1, y(v) + R + 1]$ in the unsigned coordinates.

Guard a good target whenever its connector band contains a cut position. At a fixed cell, a hop of excursion $> 2H$ uses at least $4H + 2$ transitions; their interiors are disjoint. This guard therefore costs at most $k(6s+3)\lfloor (U-1)/(4H+2) \rfloor |B|$.

There is one further root bill. Let R_B be the source vertices reaching CL_h while avoiding B. On each B-free spine interval it is an upper terminal segment, so it has at most $|B| + 1$ positive activations. At an activation c, a delayed ordinal satisfies $\alpha_i < c \leq q(u_i)$. Thus $i < O(c)$, while the checked pre-step FIFO bound gives $O(c) \leq O(q(u_i)) \leq i + 2s - 1$. There are at most $2s - 1$ such ordinals. Delete their matched simulator roots. Any remaining simulator path to a surviving crossing target projects to a source path avoiding B and continues along its connector. Its projected root is in R_B, whereas its separated actual source root is outside R_B; the activation guard forbids

this mismatch. Thus a source separator lifts to one of size at most $A|B| + F$, where

$$A = 4B_0 + k(6s+1) + k(6s+3)\lfloor (U-1)/(4H+2) \rfloor + 2s - 1,$$
$$F = 4kK_0^2(2h + 6s + 3) + 2s - 1.$$

This exact finite lift is `TransferCrossing.native_crossing_cut_lift`.

Finally use the successful-subset trace bound of [4] and the correct-subset expansion theorem of [3]. Its permitted entropy budget depends only on tape count, the simulator time coefficient and α, so it can be chosen before the fixed block size. On every remaining heavy successful fibre, the latter theorem gives a fixed $H = \lfloor \theta P \rfloor$ and cut lower bound $e|V|$ simultaneously for all $|V| \geq \alpha P$, as checked in `TransferCrossing.native_crossing_fibre_bound`. Since $U = O(P)$ and $\theta P \leq 2H$ at large P, one fixed K_0 bounds the epoch counts and one fixed $\overline{A}$ bounds A. Choose $h = \lfloor \lambda P \rfloor$ with $0 < \lambda \leq \theta$ sufficiently small that $F \leq e\alpha P/2$ at large d. Then $e|V| \leq A|B| + F$ gives $|B| \geq e|V|/(2\overline{A})$. The simultaneous choices and bounds are proved in `TransferCrossing.exists_uniform_crossing_budgets`. The simulator has at most $2^{\eta P}$ attained successful traces after choosing its fixed block size, and the fibres below the required $2^{(1-2\eta)P}$ threshold together contain fewer than $2^{(1-\eta)P}$ inputs. Every cut and output set is covered on the same remaining event. This proves the asserted quantifiers and the ambient bound. □

Theorem E concerns arbitrary macroscopic output sets at canonical widths; the direct off-carrier theorem concerns its selected slab demands at all widths. Neither subsumes the other. Theorem Q adds a different certificate: a cheap cut would need an original entry relation with almost full storage capacity. Finding a cut and a nonvacuous deficit together remains open.

I Historical feedback through actual halt

This appendix measures a different property of an original execution. Let $B = db$ and $P = dB$. Fix the native machine, a linear budget $C(P+1)$, and work-window widths $w_h \geq 2$, independent of d, b. Fix also an enlarged finite alphabet containing the source symbols and distinct spool markers and bits, with one κ-bit encoding taking blank to zero. Use this *same* encoding throughout. Each encoded B-cell work block has κ binary planes of declared capacity B bits, even when a plane happens to be constant.

The raw monitor initially retains blocks $-1, 0$; work monitor h retains blocks $-(w_h - 1), \ldots, 0$. Admit raw row zero initially. After each native step process a raw exit first, then work exits in fixed tape order. An exit shifts that window by one block. A raw shift admits the valid entering row or synthesizes exterior markers. A work shift writes every plane of the leaving block before reading every plane of the entering block. Overlapping blocks and partial writes remain in the resident window. Include pending exits after the successful transition entering the halting configuration; there is no terminal flush in the original statistic.

A write creates a new *date* at its fixed address (tape, block, plane), even when it repeats the same value. A read sees the latest preceding date, or its initial constant. The current raw-row context starts at zero and changes only when a valid raw row is admitted; exterior shifts retain it. For each produced date imported at least once before the original run ends, let ℓ_v be the maximum minus minimum of the contexts at its writer and all its importing readers. Keep dates subsequently overwritten. Set

$$\Phi(X) = \sum_v B\ell_v, \qquad W(X) = \sum_v B.$$

Repeated readers of one date share its span; a copy write creates another date. This is a statistic of the specified monitor and encoding, not a minimum over recodings or an entropy of its values.

We first prove the block-information estimate needed for the native adapter. All entropy in this appendix is in bits.

Lemma I.1 (One partition for a complete producer and consumer)**.** Lean: `feedback_global_matrix_bound`, `feedback_global_product_bound` [`FeedbackGlobalProduct.lean`]

Consider a fixed finite word of block operations. A producer begins with constant memory, has at most S bits of complete resident state, reads raw rows and previously written block versions, and emits no transpose output. Every local map has one deterministic completion on its entire available-data domain. A subsequent read-only consumer starts at the true handoff, has at most M bits of complete state, reads only raw rows and surviving final archive blocks, and emits all d correct columns in a fixed order at designated positions of the operation word, on one law of input entropy $P - \Delta$. It has no readable scratch or output readback. Let Q_r count all consumer reads, V be the total variation of the producer's raw-row admission sequence, and $\Phi = \sum_v c_v \ell_v$, where c_v is the declared capacity of each produced version read during production and its span uses the writer and all those readers. Then for every integer $1 \le g, w \le d$,

$$P \le \frac{SV + \Phi}{g} + \lceil d/w \rceil (M + \Delta) + Q_r g w b,$$

and consequently

$$\max(1, Q_r/d) \left(1 + \frac{SV + \Phi}{P}\right) \left(1 + \frac{M + \Delta}{B}\right) \ge d/72.$$

All controller, address, clock, buffer and pending fields belong to the stated resident capacities.

Outline of the formal proof. Partition raw row indices into shifted width-g intervals. Assign every producer operation the group of its current context, updating the context before a raw admission. For a uniformly chosen shift, two integer contexts at distance h have different groups with probability $\min(1, h/g)$: if $h < g$ exactly h residues place a boundary between them, and otherwise every residue does. A produced version is read by an owner different from its writer precisely when its extreme participating contexts have different groups. Thus one shift simultaneously gives

$$Ss + \sum_{v \in E} c_v \le S \sum_j \min(1, |i_{j+1} - i_j|/g) + \sum_v c_v \min(1, \ell_v/g) \le (SV + \Phi)/g,$$

where s is the number of owner switches and E the distinct produced versions used by a foreign owner. Both charges use the same shift.

Let Z contain all complete switch states and those literal foreign versions. Its full-cube fibres are Cartesian products over owner groups. Here is the precise reconstruction proving that assertion. For an attained z, replay all segments owned by a group in chronological order from its own raw rows and z. Start the first global segment in constant state; other segments use their prescribed entrance state. A read uses its own raw row, an initial constant, a foreign version in z, or an earlier own version retained in this mathematical replay. Require each segment's exit state to match the prescribed switch state, and each exported version to match its entry in z. These are tests only on that group's raw rows. Every actual input with transcript z passes them. Conversely, if every group passes, induction through the fixed global word shows that raw, own-version and foreign-version reads all agree: a foreign writer has already passed its export test. Hence the actual transcript is z. This proves the product-fibre assertion. The same replay gives one group-local function for each surviving final version, common to all its later reads. The proof-side retention of old versions adds no runtime state.

On the one actual law let $\mathcal{T} = \sum_j H(X_{I_j} \mid Z) - H(X \mid Z)$. Then $0 \le \mathcal{T} \le \Delta$. Indeed the uniform full-cube law, conditioned on z, is the product of uniform laws on the just-proved group fibres. Decomposing relative entropy with respect to that law gives

$$\Delta = D(\mu_Z \| U_Z) + \mathcal{T} + \sum_j \mathbb{E}_z D(\mu_{I_j|z} \| U_{I_j,z}) \ge \mathcal{T}.$$

All reference laws are positive on the actual support. This neither assumes the actual conditional law is a product nor asserts correctness of mixed inputs. Also $H(Z) \le Ss + \sum_{v \in E} c_v$.

Divide the d designated final column writes into $m = \lceil d/w \rceil$ chronological windows of at most w columns, the first starting at actual consumer entry. In window a, let J_a be its columns and q_a its reads, and let H_a be the owner groups of queried records. There are at most q_a such groups. Given Z, their complete raw rows and the actual M-bit entry state, replay recovers every window output, because every queried version has the same group-local function. The entropy of the unqueried groups' output rectangle conditional on those raw rows and Z is therefore at most M. Total correlation cannot increase under separate projections of the groups; subtracting the nonnegative correlation of the queried groups gives

$$\sum_j H(X_{I_j \times J_a} \mid Z) \le M + \mathcal{T} + g|J_a|bq_a.$$

Summing over windows and using conditional subadditivity on each group's partition into output columns gives

$$gb \sum_a |J_a| q_a \ge P - \Delta - H(Z) - mM - (m-1)\mathcal{T}.$$

Now $|J_a| \le w$, $\sum_a q_a \le Q_r$ and $\mathcal{T} \le \Delta$ prove the finite inequality. The transcript is paid once, each real window-entry state separately, and the law never changes.

For the product put $a = (SV + \Phi)/P$, $y = (M + \Delta)/B$, $z = 1 + y$ and $q_0 = \max(1, Q_r/d)$. Choose $g = \max(1, \lceil 4a \rceil)$ and $w = \lceil 8z \rceil$. If $g > d$ then $4a > d$, and if $w > d$ then $8z > d$; either already gives the product bound. Otherwise division by B yields $d \le ad/g + \lceil d/w \rceil y + (Q_r/d)gw$. The first term is at most $d/4$ (zero if $a = 0$). The second is at most $d/8 + z \le d/4$, since $w \le d$ implies $z \le d/8$. Thus $d/2 \le q_0 g w \le 36 q_0 (1 + a) z$, as required. The formal proof starts from one producer and one complete consumer word. It constructs the common shifted partition, actual transcript, window-entry states, slices of the consumer execution, and total read count. The final integer optimization retains the constant 72. Lean allows any bounded handoff-state map and any dated producer version in the consumer's queries, so the stated true-handoff and surviving-version restrictions are special cases. Retaining old versions in the proof does not assert a free physical archive. □

Theorem I.2 (Whole-run historical feedback)**.** Not formalized; a complete proof is given below.
For the fixed native machine, linear budget and monitor there is $c_* > 0$ such that, for every $d, b \ge 1$, $\varphi \ge 0$ and nonempty set F of inputs correctly transposed within the budget with $\Phi \le \varphi$,

$$\left(1 + \frac{\varphi}{P}\right)\left(1 + \frac{P - \log_2 |F|}{B}\right) \ge c_* d.$$

There is no original phase, read-only-work or alignment assumption.

Proof (not formalized). First postpone the complete output by a paid spool. Add one initially blank tape with a width-two monitor. Write a start marker, then simulate the original machine, writing each appended bit to the next spool cell instead of real output. Preserve all original heads and

symbols in any extra spool microstep. Continue through the actual original halt, including original work after its last output and pending monitor events. Write an end marker, then declare a phase change and flush every retained work window. Only after this phase rewind and read the spool, appending its bits to real output. Original tapes remain parked. A conservative step bound is

$$T_{\mathrm{wrap}} \le 2T + 4P + 10 \le \overline{C}(P+1), \qquad \overline{C} = 2C + 10.$$

Markers avoid an uncharged length scan or counter.

On original tapes the extra steps only insert pauses. Their raw/work shift order, per-address dates, values, partial writes and raw contexts agree exactly with the original monitor. The spool moves only right or stays in production, so its entering blocks are virgin and none of its produced versions is read in production. The terminal flush likewise has no later production reader. The old participating dates and spans therefore agree exactly: $\Phi_{\mathrm{wrap}} = \Phi$. The artificial phase is before rewind; spool rereads are consumer reads, not missing feedback.

The monitor realizes a complete block program. Outside each retained window the block store equals the native tape; inside it the resident window is authoritative. Write-before-load in either direction, retaining overlap, preserves this invariant even with partial overwrites. Serialize all κ outgoing planes before incoming ones. At phase change the flush makes the whole final archive exact; recovery is read-only. Count all windows, incoming plane tuples, unfinished payload, source control, positions, offsets, clock, pending exits and an output-column buffer. Fixed constants p_0, m_0 bound complete producer and consumer state by $p_0 B, m_0 B$: there are fixed many B-cell buffers and integer registers of $O(\log(2P))$ bits, and $\log_2(2P) \le 3B$ for $d, b \ge 1$. There is no free state reset at the phase.

A width-at-least-two monitor needs at least B head moves before its first exit and between successive exits, in either direction. With $K = k + 1$ work tapes including the spool, valid raw admissions have total row variation $V \le v_0 d$, $v_0 = 4\overline{C}$, and all recovery reads obey $Q_r \le q_0 d$, $q_0 = 2\overline{C}(1 + \kappa K)$. To see the former, entering raw blocks change their valid indices by at most two per raw shift; exterior shifts cannot increase the variation of the valid subsequence. The latter follows by bounding every raw/work exit by $\lfloor T_{\mathrm{wrap}}/B \rfloor \le 2\overline{C}d$, counting all planes. Likewise original participating capacity satisfies $W \le u_0 P$, $u_0 = \max(1, 2\overline{C}\kappa k)$, since every such date is exported at an original work shift. Terminal flush versions and initial constants do not participate.

Fix a coarse event word recording raw/work shifts with tape and direction, the phase, and complete output-column flushes. Its alphabet has $2K + 4$ letters; its length is at most $[(2\overline{C}(K + 1) + 2)d]$. Such a word fixes all addresses, latest-date ties, raw contexts and phase labels; flushes expand into fixed lists of planes. Thus there are at most $2^{C_e d}$ words, where

$$C_e = [2\overline{C}(K+1) + 2] \log_2(2K + 5).$$

One nonempty fibre $F_0 \subseteq F$ has $|F_0| \ge |F| 2^{-C_e d}$, hence $\Delta = P - \log_2 |F_0| \le P - \log_2 |F| + C_e d$. Use its one uniform law throughout.

The fixed-word block maps have common local full-cube completions. Between prescribed I/Os simulate from the complete resident tuple with the bounded clock, stopping before any unexpected external observation. Invalid local codes, wrong events, premature halt or timeout give fixed local defaults for the remaining prescribed stages. Previously imported values are resident data; the next prescribed read is the only new external value. All guards and clocks are already charged. On F_0 every guard follows the actual wrapped computation; off it, correctness and native validity are unnecessary. No support-membership or global input test is used.

Apply Lemma I.1. It gives

$$\max(1, q_0)(1 + p_0 v_0 + \varphi/P)(1 + m_0 + (P - \log_2 |F| + C_e d)/B) \ge d/72.$$

Since $d/B = 1/b \le 1$, take

$$c_* = [72 \max(1, q_0)(1 + p_0 v_0)(1 + m_0 + C_e)]^{-1}.$$

The displayed inequality implies the result by bounding each parenthesis above by its constant factor times the corresponding parenthesis in the statement. This proves the claim for every nonempty F and every width. □

Corollary I.3 (Success tails and many long-span dates)**.** Lean (partial): `FeedbackTail.success_count`, `FeedbackTail.fixed_linear_count`, `FeedbackTail.linear_in_dimension_probability`, `FeedbackTail.many_long_span_dates` [`FeedbackTail.lean`]

Not formalized: the native monitor/spool theorem and its declared-capacity budget. Given those inequalities, Lean proves the exact counts, the ambient probability bound and the finite dated-plane conclusion.

Let E_φ be the timely correct inputs with $\Phi \le \varphi$. Then, including the empty case,

$$|E_\varphi| \le 2^{P - B[c_* d/(1+\varphi/P)-1]_+}.$$

For every fixed $s \ge 0$, $|E_{sP}| \le 2^{(1-\eta_s)P}$ at large d, with $\eta_s = c_*/[2(1+s)]$. For $\alpha = c_*/4$, $|E_{\alpha P d}|/2^P \le 2^{-B}$ at large d. Every other timely correct execution has more than $\alpha d/2$ dated B-bit planes spanning at least $\alpha d/(2u_0)$ raw-row contexts.

Proof (not formalized). Rearrange Theorem I.2 and use nonnegative source deficit. For $\varphi = sP$, take $d \ge 2(1+s)/c_*$. For $\varphi = \alpha P d$ and $d \ge 1/\alpha$, the quotient $c_* d/(1+\alpha d)$ is at least two, giving deficit at least B. For an execution with $\Phi > \alpha P d$, let H_* be the total declared capacity of dates with span at least βd, $\beta = \alpha/(2u_0)$. Since all spans are at most $d-1$ and $W \le u_0 P$, $\Phi \le \beta d W + d H_* \le (\alpha/2) P d + d H_*$. Thus $H_* > (\alpha/2)P$, which is more than $\alpha d/2$ planes. □

The probability statement is an ambient joint success/low-feedback bound; under full correctness it is a probability over the whole input cube. Different dates at the same cell may contribute separately. Large feedback is itself cheap: copy the raw rows in order, park the raw context at $d-1$, then rescan the copy. On a tape with fixed window width w, block j is exported when the aligned copying heads first enter block $j+w$. For $0 \le j \le d-w-2$, its produced date has writer context $j+w$ and is imported during the rescan at context $d-1$. All κ planes count, including constant ones. Consequently, for $d > w+1$,

$$\Phi \ge \kappa B \sum_{j=0}^{d-w-2} (d-1-j-w) = \frac{\kappa B (d-w-1)(d-w)}{2}.$$

The declared capacity of all participating dates is $O(P)$ and every span is at most $d-1$, so $\Phi = \Theta(Pd)$ in $O(P)$ native time. The reads may feed a parity accumulator without changing this statistic. Hence neither the feedback nor its long dates are a time bill or independent useful information.

J Formalization boundary and verification

The library imports the earlier native-relations package with the repository's pinned Lean/Mathlib toolchain. It reuses the frozen `Transpose.TM` model, the published batch simulator and the shared `lean-machine` library. No frozen statement is changed. Each numbered result names its checked endpoint or states the remaining gap; the generated index lists the declarations.

Actual native replay and counting. The adaptive replay proof starts from actual native execution prefixes. Chronological write dictionaries recover work reads from their true latest parents. Comparing two runs with the same paid record recovers their states, positions, old tuples and output ordinals, then proves input injectivity. This comparison needs no total decoder on invalid messages. The finite metadata and sparse support/value menus are counted separately. Neither a common fine trace nor a suitable selected chunk is assumed.

The return theorem derives its slab from each timely correct run, retains the snapshot-adjusted geometric margin, and proves its uniform large-dimension tail for all positive entry widths. The input-cut theorem uses literal endpoint-inclusive separators in the original dependency graph. It constructs the cut-free ancestors, derives their state and latest-parent closure, and recovers the input from the actual fragment record. Its sparse-menu estimate handles every cut budget without a half-runtime restriction. The localized variant derives the snapshot overlays and exterior-parent coverage from each actual selected slab. Its final off-carrier expansion bound controls every valid slab and every free head, with targets in the complete original run. The dimension thresholds precede the width, machine and input quantifiers. The coherent-cluster extension proves multi-core replay and constructs its finite metadata from actual signed charts. Only the rounded endpoints are counted, not descriptions of real functions. Its return and cut tails hold simultaneously for all clusters, signs and charts; the cut statement covers every valid selected slab and all smaller excursion thresholds.

The getter replay includes actual observer updates, continuous port positions, the shared label, strict prior output and joint sparse corrections. Bounded metadata is extracted from actual runs. The paid scan loader separately implements its transition table on the frozen native model: it takes exactly $2P + 4$ steps, has the stated finite state and alphabet cardinalities, and continues the arbitrary consumer without a linking step. For a consumer that preserves each archive symbol it reads, the reached-configuration invariant and shared scan checkpoints construct total local getters with zero errors throughout preparation and consumption. The resulting success bound is formalized, including the case of no scan archives.

The complete block-program lemma. The producer is an actual total causal program. One shifted partition pays both its switch states and distinct foreign versions. Chronological replay proves full-cube product transcript fibres and group-local recovery; the entropy proof applies to one arbitrary input law, without independence. The consumer is one complete read-only operation word. Its window entry states and output records are obtained by slicing that execution, and its window read counts sum to at most the actual global count. Thus Lemma I.1, including the constant 72, is formalized. The theorem does not itself compile the native tape monitor or output spool.

Reusable finite components. The shared library supplies monotone sample and signed-position menus, finite-field hash separation, common context pruning, actual top sets, weighted selection, strict small-fibre bounds, time grids and bounded-degree tape-word graph balls. The persistence selection uses these actual top sets under the original law to choose one tape set, one live interval and one retained event. Its conditional bounds concern the whole native tape words: blank extension identifies the finite horizon word with the infinite word. Native correctness bounds the actual strict-prefix fibre and hence the number of bits still undelivered. The uniform-source wrappers derive their half-mass escape events from the literal native thin-source count, including its factor two where required. Every subsequent selection, probability and uniform rate estimate is checked. For remote persistence, one common finite observation window is used while each input retains its own first halt time; no frozen post-halt configuration counts as a future read. The native catalogue theorems

remain explicit premises. The numerical feedback tails and finite weighted-date conclusions are likewise checked from the stated native monitor inequalities. The carrier/reference/cache corollaries derive their complete state and label budgets from the literal finite products, including ceilings, default states and every syndrome. Their common native simulation contracts remain explicit. The reference and cache tails retain the common-hash factor two. Actual unchanged-cell intervals and changed-address duration/span bounds are also checked, including episodes truncated at the successful halt horizon.

Other checked components include the continuous order-conflict chart, protected cores and paid fringes, actual changed-cell supports, shared scan checkpoints, joint entropy invoices and input erosion. The frozen-trace cut invoice derives input recovery from the native field recursion and the literal separator; its field-only form pays an output bit only at cut desired roots. This information step does not establish a geometric simulator lift.

The actual aligned simulator's batch clock, terminal drain, head shadows and FIFO bounds are checked. The guarded-path theorem constructs and counts literal simulator vertices, and projects surviving paths through actual source state and last-visit edges. It includes both path endpoints and needs no shared source trace. Raw-cone transport is also checked, including the version with no terminal-output charge when the future ordinal exists. The finite eroded-input invoice combines an actual lifted cut with the full native frozen-trace injection. The entry-capacity theorem extracts an actual nonlinear storage code from a joint guard/exterior fibre; its source relation retains all lanes and self-edges. One common simulator-trace event controls every original capacity certificate, with no union over their parameters. Crossing-target transport is also checked: literal epoch labels bound the discarded targets, actual source crossings supply the surviving connectors, and long-hop packing pays the target guards. The finite source-cut lift is combined with the correct-subset crossing theorem of the earlier paper and uniform budgets to derive expansion on the original executions.

Remaining execution-level obligations. The dynamic-carrier theorem requires a new transition-table compiler, a physical visited-interval invariant through arbitrary handoffs, partial sparse decoding and an exact state count. The existing batch simulator is a different machine. The catalogue and raw-relative handoffs, historical-cache policy, last-visit eraser and redundant-archive wrapper also need their stated native constructions. Consequences of those constructions retain the same explicit dependency.

The whole-run feedback theorem requires a native block monitor and paid output spool through actual halt, with the complete state and event budgets and preservation of the dated-version statistic. The native constructions remain formalization todos. The simulator transport adapters are complete over the existing aligned machine. A finite helper is never substituted for a missing execution correspondence.

Every cited Lean declaration must use only `propext`, `Classical.choice` and `Quot.sound`, as checked by the generated axiom audit. No proof is admitted, and no additional axiom or kernel-bypassing computation is used. The English outlines of formalized results are reading aids; their Lean statements and proofs are authoritative. This paper is not a fully formalized proof of every native theorem it contains.

K Index of Lean declarations

The following declarations are cited by formal-status lines. Names are relative to `AdaptiveTransfer`. A partial status retains the gap stated at its theorem.

ClusterExpansion.lean native_coherent_cluster_expansion, native_coherent_cluster_returns.

ContextPruning.lean context_conditional_bound, context_half_mass, context_subset.

CurrentVersion.lean ExecutionPrefix.filtered_native_read, filtered_read_correct.

EnlargedFinite.lean Enlarged.finite_bound.

FeedbackGlobalProduct.lean feedback_global_matrix_bound, feedback_global_product_bound.

FeedbackTail.lean FeedbackTail.fixed_linear_count, FeedbackTail.linear_in_dimension_probability, FeedbackTail.many_long_span_dates, FeedbackTail.success_count.

Fragments.lean fragment_count_le, localized_fragment_count_le.

FrozenCutInvoice.lean native_cut_recovers, native_frozen_cut_entropy_lower, native_frozen_field_invoice, native_frozen_full_invoice.

GetterFinite.lean Getter.native_finite_getter_bound.

GetterGeometry.lean Getter.native_fixed_geometry_bits_bound.

HistoricalCache.lean ExecutionPrefix.changedCells_card_le_duration, ExecutionPrefix.residual_card_le, ExecutionPrefix.residual_card_le_entering_add_episode, ExecutionPrefix.unchanged_outside_changedCells.

IncomingReturns.lean outside_first_read_is_long_return.

InformationAccounting.lean joint_record_card_bound, joint_record_entropy_identity, joint_record_information_lower.

InputCutTail.lean native_input_cut_tail, native_input_cut_tail_all_width.

InputErosion.lean demanded_erosion_loss, interval_union_erosion_loss.

NativeImportInformation.lean NativeReplayClass.code_injective, NativeReplayFamily.native_import_information_saving.

NativeInputCutEndpoint.lean mem_nativeBadInputCuts_iff, native_bad_input_cut_bound.

NativeReturnCounting.lean native_finite_return_bound, native_positive_saving_observation.

NativeWrites.lean instruction_eq_of_old_tuple, step_preserves_unscanned, step_scanned_write, step_work_update.

OffCarrierExpansion.lean native_off_carrier_expansion.

OrderChart.lean finite_order_chart, order_conflict_le_two_error.

PersistenceFutureSource.lean PersistenceFutureSource.of_thin_source_bound.

PersistenceSource.lean ConditionalPersistence.of_small_thin_source.

PrefixFinite.lean Enlarged.prefix_finite_bound.

ProtectedCores.lean normalized_chart_core_fringe_counts, outside_avoids_core, protected_core_disjoint_fringe, selected_in_core_or_fringe.

RecordCounts.lean replay_message_bound, sizedImports_card, sparseImports_card.

ReferenceTail.lean cache_tail_of_half_simulation, reference_cache_sparse_reserves, reference_tail_of_half_simulation.

ReplayGeometry.lean integerSlab_card, native_fixed_geometry_replay_bound.

ReturnTail.lean native_macroscopic_return_tail, returnTailEpsilon_pos, returnTailEta_pos.

ScanCheckpoints.lean checkpointLabel_card, checkpoint_image_card_le, checkpoint_residue_average, preceding_residue_anchor, scanState_window.

ScanLoader.lean ScanLoader.native_load, ScanLoader.native_ready_properties, ScanLoader.native_resources.

SlabGeometry.lean exists_slab_geometry, native_slab_geometry.

Storage.lean ExecutionPrefix.persistent_cells, ExecutionPrefix.untouched_cell_constant, native_support_growth_le_one, native_support_shrink_le_one.

StoresFromCarrier.lean native_support_window_complete, native_two_dense_heads, stores_tail_of_carrier.

TransferCrossingTail.lean TransferCrossing.uniform_original_crossing_expansion.

TransferEntryNative.lean TransferEntry.certified_original_cut_lower_bound, TransferEntry.native_original_entry_capacity, TransferEntry.uniform_entry_success_tail.

TransferErodedInvoice.lean TransferEroded.native_eroded_fibre_invoice, TransferEroded.native_eroded_fibre_invoice_cover.

TransferNativeCones.lean TransferCones.native_cone_transport, TransferCones.native_cone_transport_interior.

TransferNativeProjection.lean TransferGuards.native_guarded_projection.

Other declarations named in the text `ConditionalPersistence.of_half_mass_escapes`, `Enlarged.macroscopic_tail`, `Enlarged.prefix_macroscopic_tail`, `ExecutionPrefix.adaptive_replay_agreement`, `ExecutionPrefix.dense_persistent_interval`, `ExecutionPrefix.many_episode_changes_duration_span`, `ExecutionPrefix.replay_output_count_agreement`, `Getter.error_support_le_returns`, `Getter.exists_getter_density`, `Getter.native_macroscopic_getter_tail`, `NativeReplayClass.LocalCut.records_recover`, `PersistenceFutureRates.of_half_mass_escapes_all_widths`, `ScanLoader.nativeArchiveGetters_zero_errors`, `ScanLoader.native_consumer_halt`, `ScanLoader.native_prepare_then_consume`, `ScanLoader.native_scan_loader_exponential`, `ScanLoader.native_scan_loader_tail`, `SlabObservation.encode_times`, `TransferCrossing.exists_uniform_crossing_budgets`, `TransferCrossing.native_crossing_cut_lift`, `TransferCrossing.native_crossing_fibre_bound`, `TransferEroded.native_eroded_cut_lift`, `TransferFIFO.actual_ordinal_lookahead`, `TransferFIFO.actual_sourceRoot_le_clock`, `binary_hash_bits_eq_ceiling`, `binary_matrix_collision_card`, `carrier_state_bits_small`, `cluster_cut_rates`, `cluster_return_rates`, `conditional_entropy_le_mean_log_support`, `constantSlabClass`, `encodeReplay_injective`, `exists_binary_common_half`, `exists_sparse_injective_hash`, `geometry_cut_menus_logb_small`, `globalGeometryMenu_card`, `headDescription_injective`, `incoming_support_le_returns`, `joint_import_information_from_support`, `localGeometryMenu_card`, `nativeInputCone_determines`, `native_bad_cluster_cut_bound`, `native_bad_cluster_return_bound`, `native_bad_slab_cut_bound`, `native_cluster_one_head_bad`, `native_cluster_slab_geometry`, `native_cluster_two_return_heads`, `native_local_cut_tail`, `native_metadata_logb_small`, `native_one_head_bad`, `native_sparse_logb_small`, `native_two_return_heads`, `offCarrierReturns_card_eq_sum`, `replayImports_card_le`, `replayMetadata_card`, `sampledWalk`, `totalFragmentRecord_card`.

Imported declarations `LeanMachine.FiniteSourceEscape.half_mass_pair_escape`, `LeanMachine.linearHash_update`, `NativeRelations.exists_offset_slabTransitions_le`, `NativeRelations.exists_slab_chunk_dichotomy`, `NativeRelations.outIn_card_le_two_indep_add_six_runs`.

L Building and verifying

The repository must be cloned with its sibling paper directories. The Lean package pins Lean and Mathlib to version 4.34.1 and imports earlier libraries through relative paths. From the paper directory, run

```
python3 tools/lean_index.py
cd lean
lake exe cache get
lake build
lake env lean PaperAxioms.lean
cd ../paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The generator also rewrites the root imports to include every public local module. The generated index and axiom file list every declaration cited by a formal-status line and every Lean declaration explicitly named in the text. The permitted axioms are `propext`, `Classical.choice` and `Quot.sound`. There are no admitted proofs, extra axioms or native computation shortcuts. Build products under `.lake/` are local and are not part of the published sources. The README records the remaining

formalization obligations; a successful build does not certify the written-only parts. The current status lines classify the 48 numbered results as 35 fully formalized, six partially formalized and seven with complete written proofs not yet formalized. The generated index records the cited declarations; the library root includes every public local module. The verification records preserve the scope of earlier reviews separately from later formalization.

The geometric slab-selection lemma uses a local `maxHeartbeats` limit of 800000. This changes only the elaboration allowance; kernel checking and the permitted axioms are unchanged.

References

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